MBG 1201 Introduction to Modern Biology

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- Organisms are composed of matter
- Matter is anything that takes up space and has mass. Matter is made up of elements.
- An element is a substance that cannot be broken down to other substances by chemical reactions.
- A compound is a substance consisting of two or more different elements combined in a fixed ratio.
- Of the 92 natural elements, about 20–25% are essential elements that an organism needs to live a healthy life and reproduce.
- oxygen (O), carbon (C), hydrogen (H), and nitrogen (N)—make up approximately 96% of living matter.
- Calcium (Ca), phosphorus (P), potassium (K), sulfur (S), and a few other elements account for most of the remaining 4% or so of an organism's mass.
- Trace elements are required by an organism in only minute quantities.

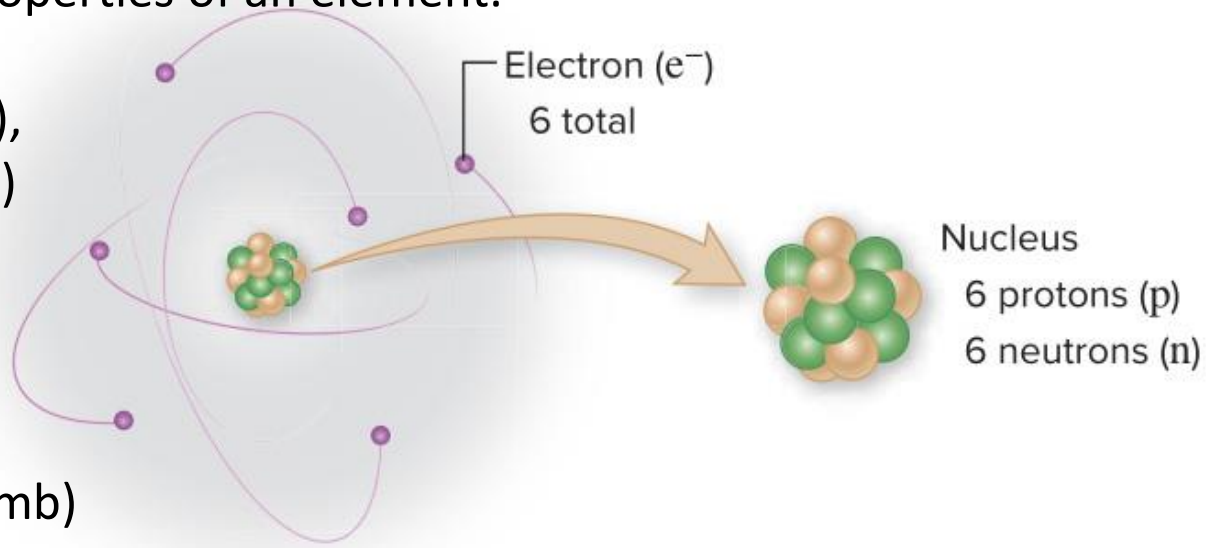
Atom : the smallest unit of matter that still retains the properties of an element.

Subatomic Particles:

Proton (p^+) : positively charged (1.6×10^{-19}Coulomb),
mass~=1.7×10^{-24}g ~=1 Dalton (or amu)

Neutron (n^0) : neutral,
mass~= 1.7×10^{-24}g

Electron (e^-): negatively charged (-1.6×10^{-19} Coulomb)
mass~= 9×10^{-28}g



protons and neutrons form the nucleus

All atoms of a particular element has the same number of protons which is unique to that element and is called the atomic number.

$^{23}_{11}Na$

Mass number = number of protons + neutrons
= 23 for sodium

Atomic number = number of protons
= number of electrons in a neutral atom
= 11 for sodium

Number of neutrons = mass number − atomic number
= 23 − 11 = 12 for sodium

For a neutral atom: Number of protons = number of electrons

Isotopes

Isotopes: Different atomic forms of the same element due to differences in the numbers of its neutrons.
Although their number of neutrons differ they behave the same in chemical reactions.
In nature, an element may occur as a mixture of its isotopes. For an element with more than one naturally occurring isotope, the atomic mass is an average of those isotopes, weighted by their abundance. Thus, carbon has an atomic mass of 12.01 daltons.
For example carbon has 15 different known isotopes from C-8 to C-22 and it is found as a mixture of its isotopes in nature.
Only C-12 and C-13 are stable (not radioactive).
A radioactive isotope is one in which the nucleus decays spontaneously, giving off particles and energy. When the radioactive decay leads to a change in the number of protons, it transforms the atom to an atom of a different element. For example, when a carbon-14 (14C) atom decays, a neutron decays into a proton, transforming the atom into a nitrogen (14N) atom.

$^{12}_{6}C$ 98.89% in nature.

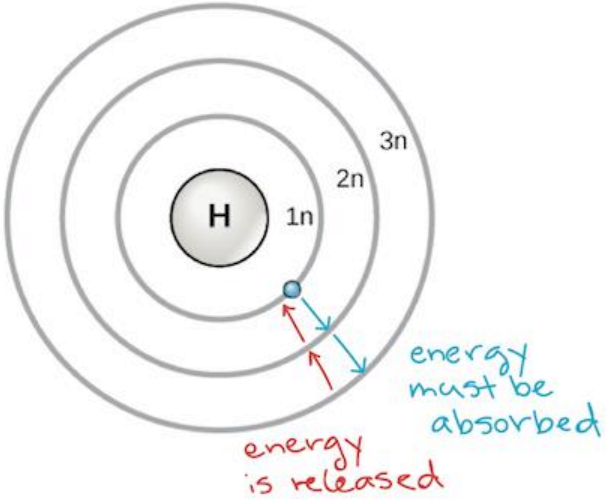
$^{13}_{6}C$ 1.11% in natüre.

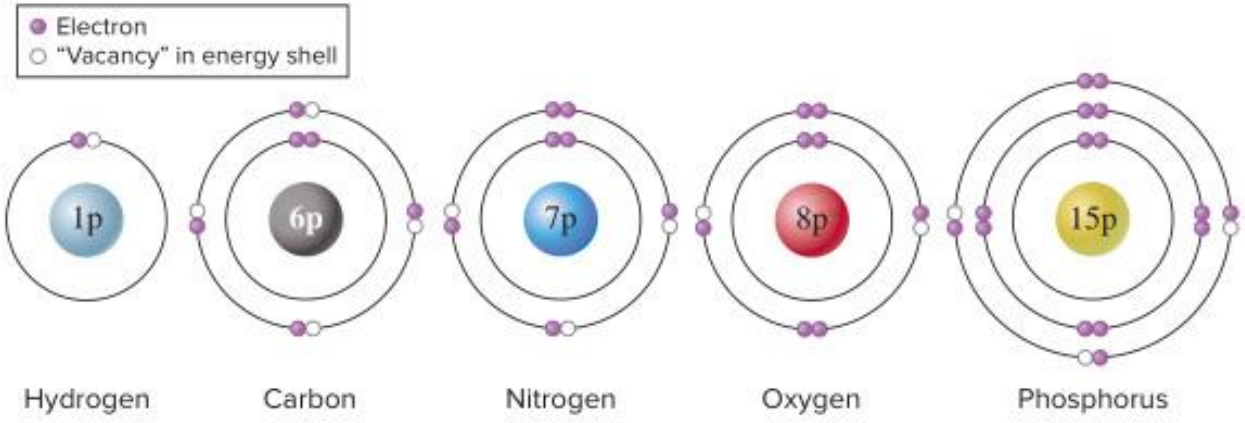
$^{14}_{6}C$ Useful for radiocarbon dating: C-14 has a half life of 5700 years. one in a trillion carbon atoms are C-14. The C-14 atoms decay and are replaced by new C-14 atoms at a constant rate. All organisms have the same percentage of C-14 in their bodies. When they die C-14 decays at a predictable rate. And the time passed over their deaths can be calculated.

The Energy Levels of Electrons

Electrons exist only at fixed levels of potential energy called electron shells.
Each electron shell has a different energy level.
An electron can exist only at certain energy levels, not between them.
An electron can move from one shell to another, but only by absorbing or losing an amount of energy equal to the difference in potential energy between its position in the old shell and that in the new shell.

The number of electrons in the outermost shell of a particular atom determines its reactivity (tendency to form chemical bonds with other atoms).
valence shell, outermost shell
valence electrons. the electrons found in the valence shell





Electron distribution diagrams.

Electron Distribution and Chemical Properties

- The chemical behavior of an atom depends mostly on the
- number of electrons in its outermost shell.

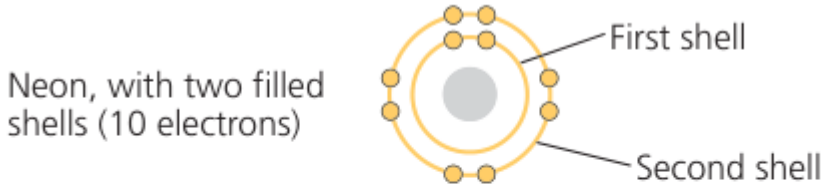
- Those outer electrons are valence electrons and the outermost electron Shell is the valence shell.

- Atoms with the same number of electrons in their valence shells exhibit similar chemical behavior.

For example, fluorine (F) and chlorine (Cl) both have 7 valence electrons, and both form compounds when combined with the element sodium (Na).

- An atom with a completed valence shell is unreactive

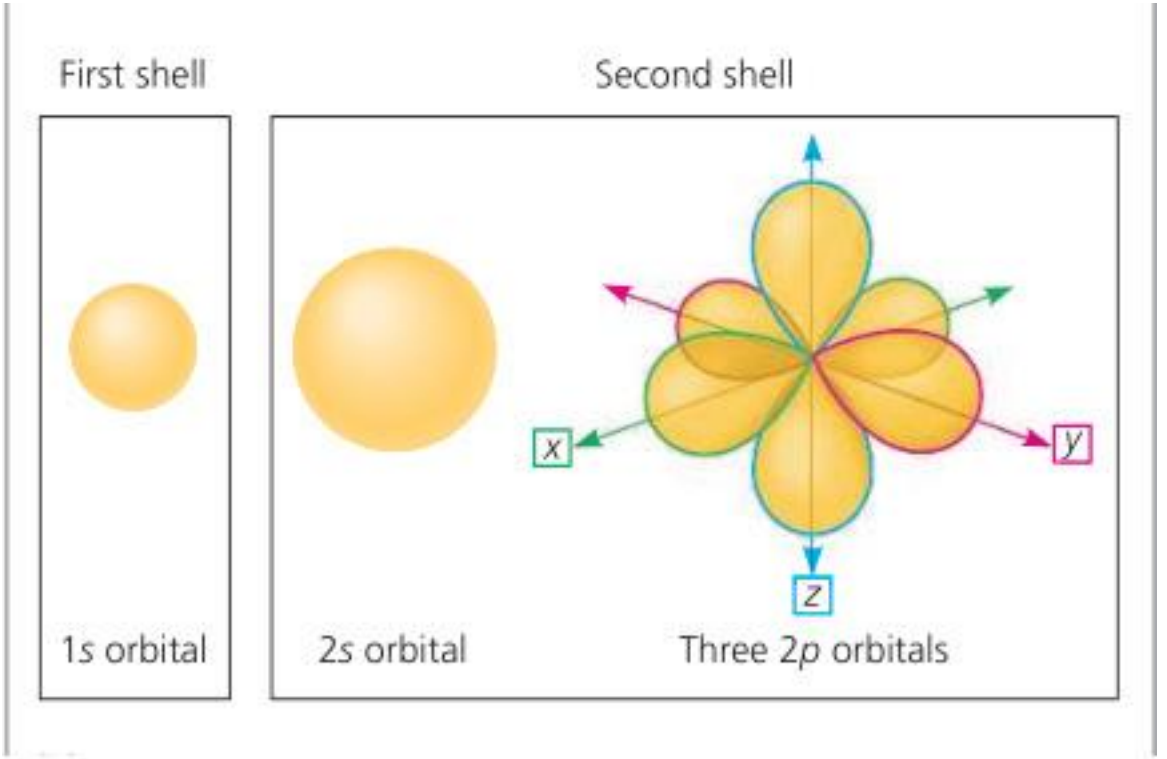
it will not interact readily with other atoms. These elements are said to be inert, meaning chemically unreactive.



(a) Electron distribution diagram. An electron distribution diagram is shown here for a neon atom, which has a total of 10 electrons. Each concentric circle represents an electron shell, which can be subdivided into electron orbitals.

Electron Orbitals

- the concentric-circle diagrams do not give a real picture of an atom. In reality, we can never know the exact location of an electron. What we can do instead is describe the space in which an electron spends most of its time. The three-dimensional space where an electron is found 90% of the time is called an orbital.
- No more than 2 electrons can occupy a single orbital.
- The reactivity of an atom arises from the presence of unpaired electrons in one or more orbitals of the atom's valence shell.



Each orbital holds a maximum of 2 electrons. The first electron shell, on the left, has one spherical (s) orbital, designated 1s. The second shell, on the right, has one larger s orbital (designated 2s for the second shell) plus three dumbbell-shaped orbitals called p orbitals (2p for the second shell). The three 2p orbitals lie at right angles to one another along imaginary x-, y-, and z-axes of the atom.



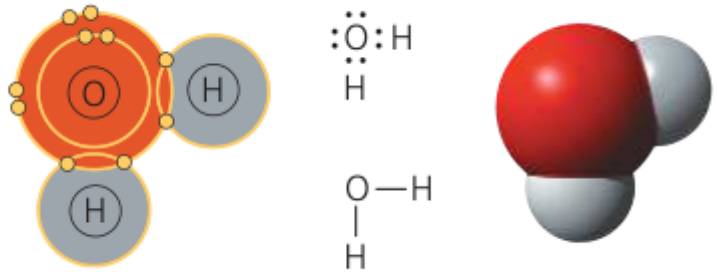
The formation and function of molecules and ionic compounds depend on chemical bonding between atoms

Atoms with incomplete valence shells can interact with certain other atoms in such a way that each partner atom completes its valence shell:
The atoms either share or transfer valence electrons.
These interactions usually result in atoms staying close together, held by attractions called chemical bonds.

The strongest bonds are covalent bonds in molecules and ionic bonds in dry ionic compounds.

Covalent Bonds
Forms when valence electrons are shared between atoms.
Two or more atoms held together by covalent bonds form molecules.
e.g. CH_4, H_2O, O_2

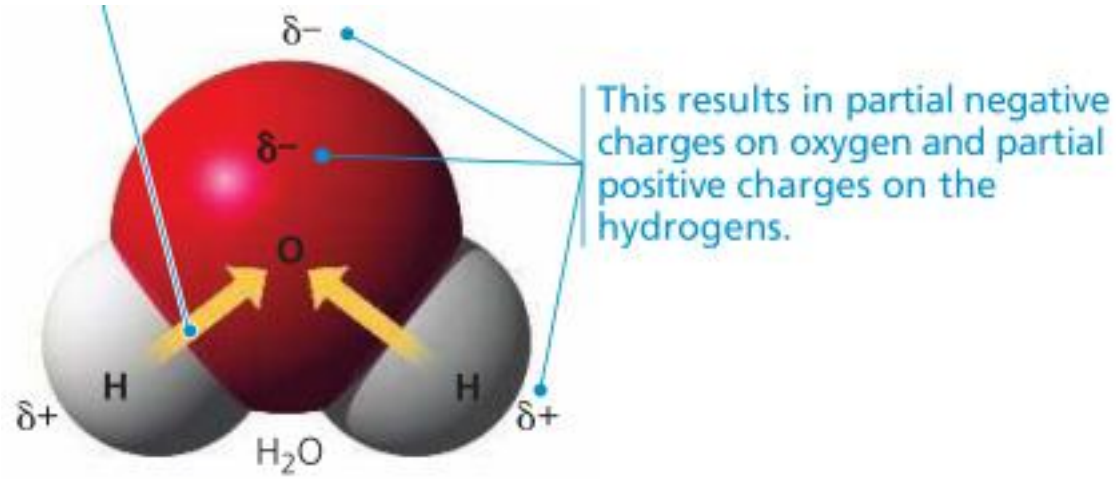
(c) Water (H_2O). Two hydrogen atoms and one oxygen atom are joined by single bonds, forming a molecule of water.



The bonding capacity is called the atom's valence and usually equals the number of electrons required to complete the atom's outermost (valence) shell.
Pure elements: H_2, O_2
Compounds: CH_4, H_2O (a compound is a combination of two or more different elements)

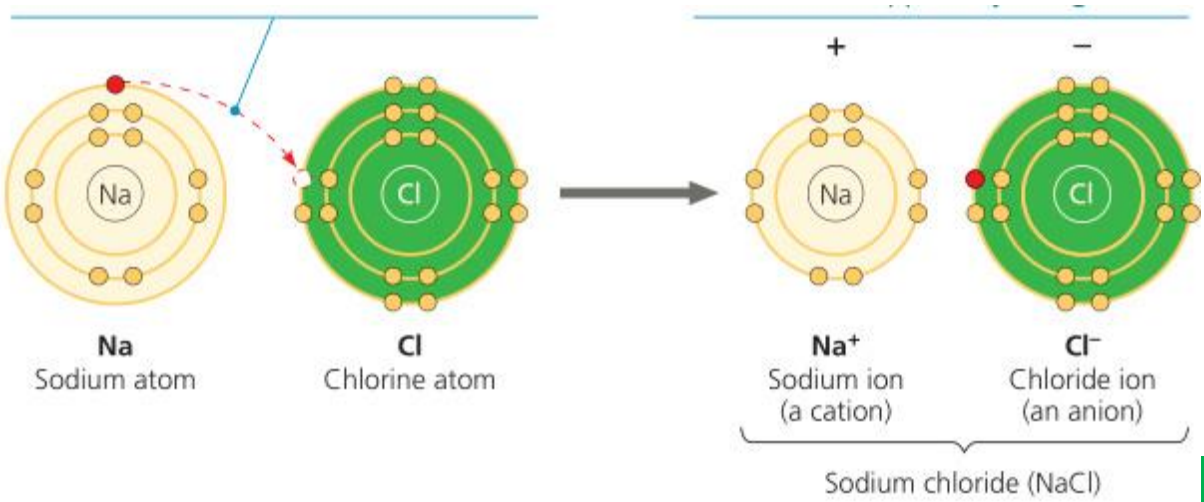
Electronegativity

Electronegativity is a chemical property that describes an atom's ability to attract and hold onto electrons when it forms a chemical bond with another atom.
it measures how strongly an atom pulls electrons towards itself in a bond.
nonpolar covalent bonds: when atoms that form a bond have equal electronegativity they share the electrons equally.
polar covalent bonds: when an atom that forms a bond is more electronegative the electrons are not shared equally resulting in partial charges on the molecules.

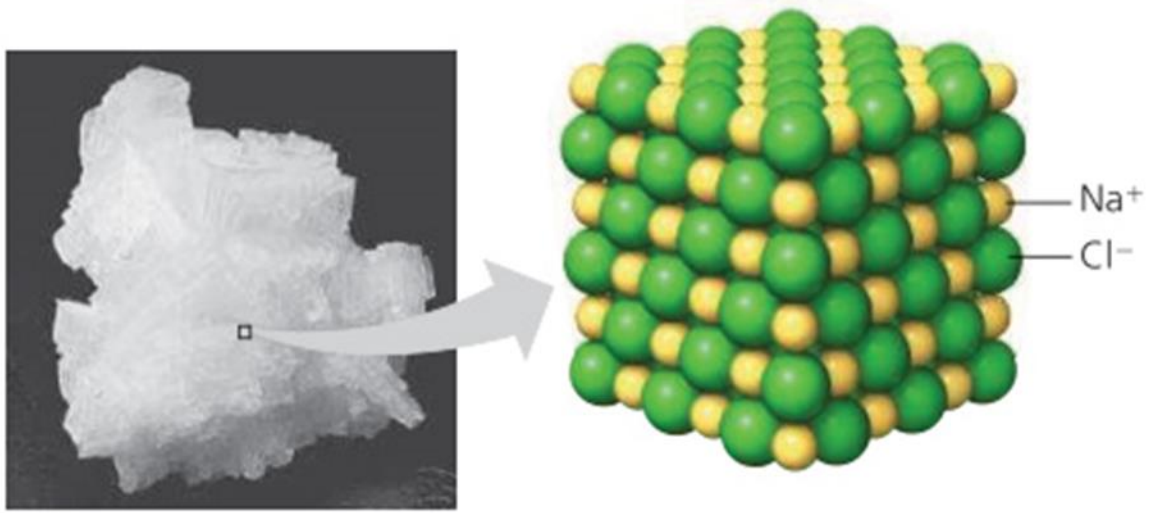


Ionic Bonds

Ionic Bond: If the electronegativity difference is very large, one atom can pull an electron away from the other entirely, forming an ionic bond (like in NaCl, where Na donates an electron to Cl).
The two resulting oppositely charged atoms (or molecules) are called ions.
Cation: positively charged ion
Anion: negatively charged ion
Opposite charges on ions attract each other resulting in an ionic bond.
Compounds formed by ionic bonds are called ionic compounds, or salts.
Ionic compounds are not composed of molecules. For example NaCl is not a molecule
the formula NaCl only represents the ratio of
Na and Cl elements in a crystal of the salt.
Electrically charged molecules are also called ions.
ammonium ($NH4^+$)



▼ **Figure 2.13 A sodium chloride (NaCl) crystal.** The sodium ions (Na^+) and chloride ions (Cl^-) are held together by ionic bonds. The formula NaCl tells us that the ratio of Na^+ to Cl^- is 1:1.



Weak Chemical Interactions

Weak Interactions

Reversible : The reversibility of weak interactions can be an advantage: Two molecules can come together, affect one another in some way, and then separate.

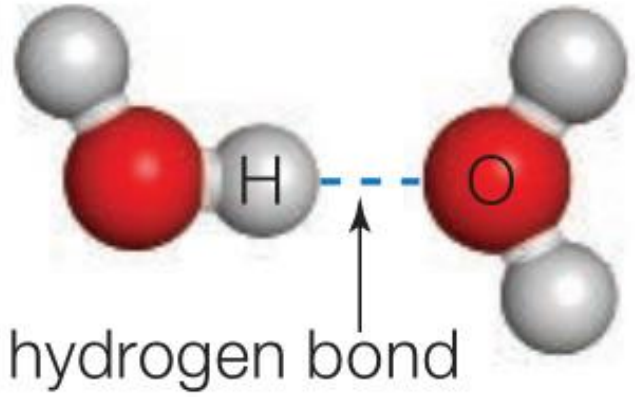
Important for living things.

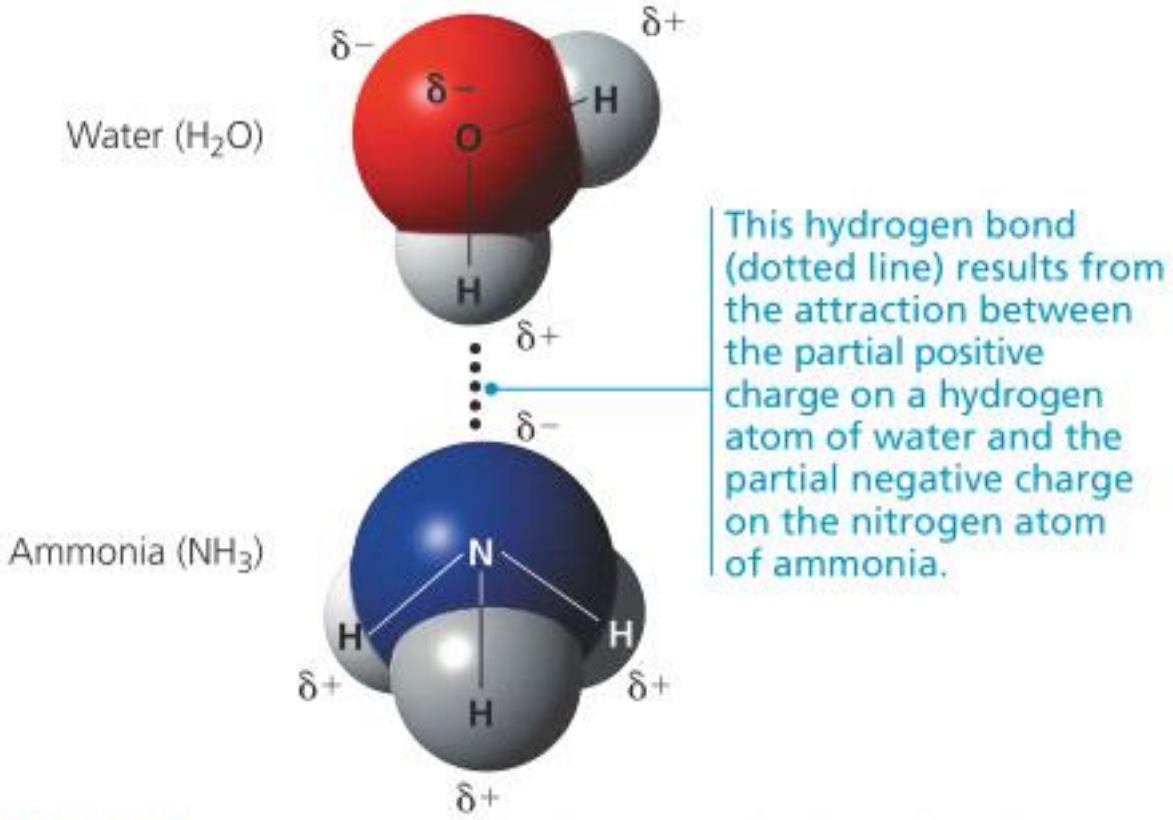
Non covalent interactions give DNA its uniqe characteristic as the carrier of genetic information.

Protein structure depends on weak interactions as well as covalently bound peptide chain.

Hydrogen Bonds

B A hydrogen bond is an attraction between a hydrogen atom and another atom taking part in a separate polar covalent bond.



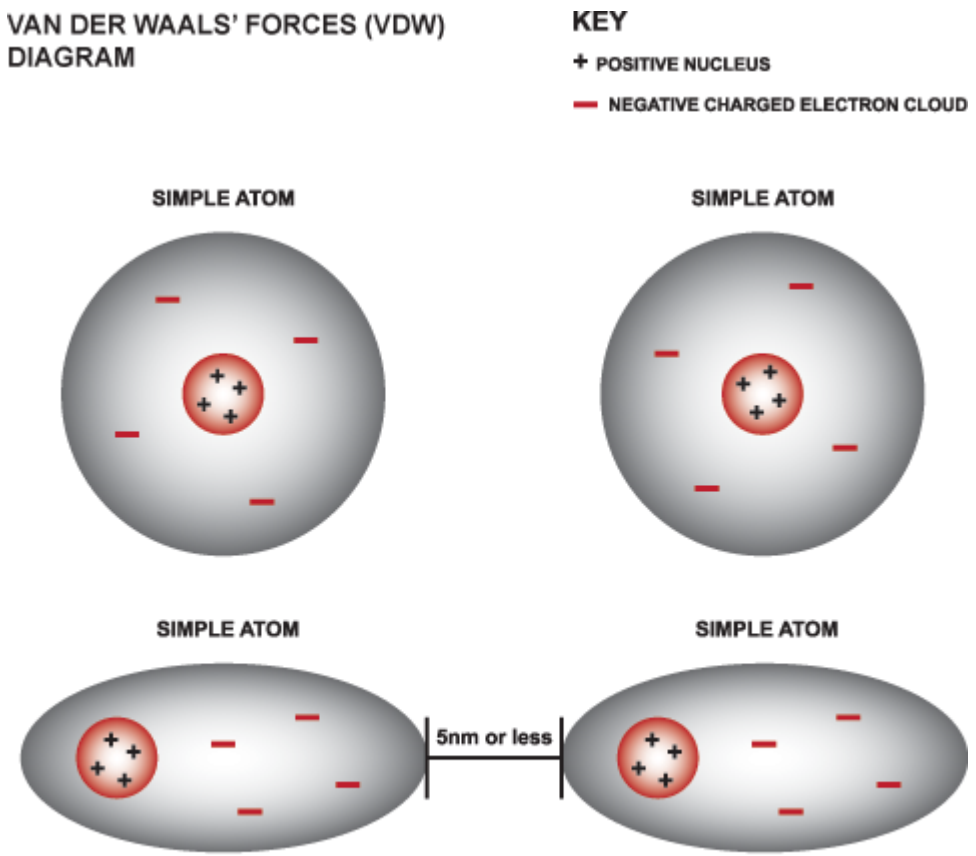


When a hydrogen atom is covalently bonded to an electronegative atom, the hydrogen atom has a partial positive charge that allows it to be attracted to a different electronegative atom with a partial negative charge nearby. This noncovalent attraction between a hydrogen and an electronegative atom is called a hydrogen bond.

In living cells, the electronegative partners are usually oxygen or nitrogen atoms.

Van der Waals Interactions

Exists in all molecules.
Predominant forces in nonpolar molecules.
In nonpolar molecules the electrons are generally distributed evenly.
Since electrons are constantly moving there are times when one side of the atom has more electrons than the other side forming more negative charge in that area. The atom is polarized in that instant causing a temporary dipole (a polarized molecule with one side positive and the opposite side negative). This polarized atom cause another nearby nonpolar atom to polarize temporarily (induced dipole).
For nonpolar molecules as electron number and MW increases polaribility increases.



When two atoms come within 5 nanometers of each other, there will be a slight interaction between them, thus causing polarity and a slight attraction.

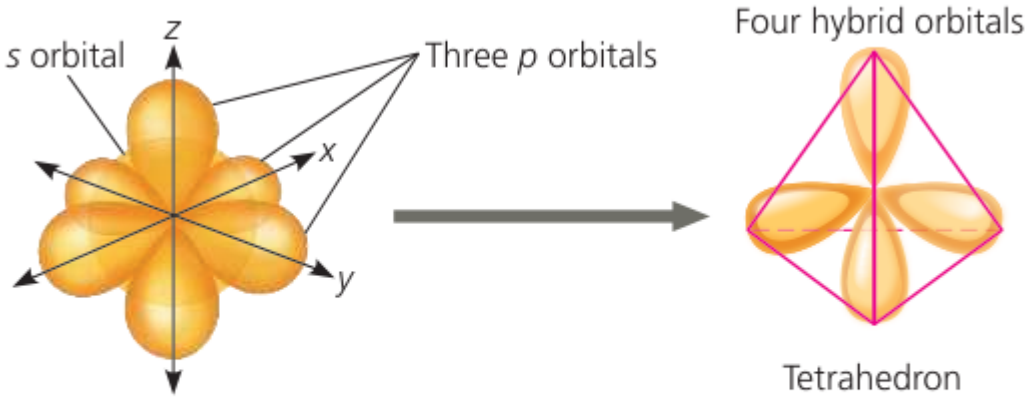
Molecular shape is key to function

A molecule has a characteristic size and shape, which are key to its function in the living cell. A molecule consisting of two atoms, such as H2 or O2, is always linear, but most molecules with more than two atoms have more complicated shapes.

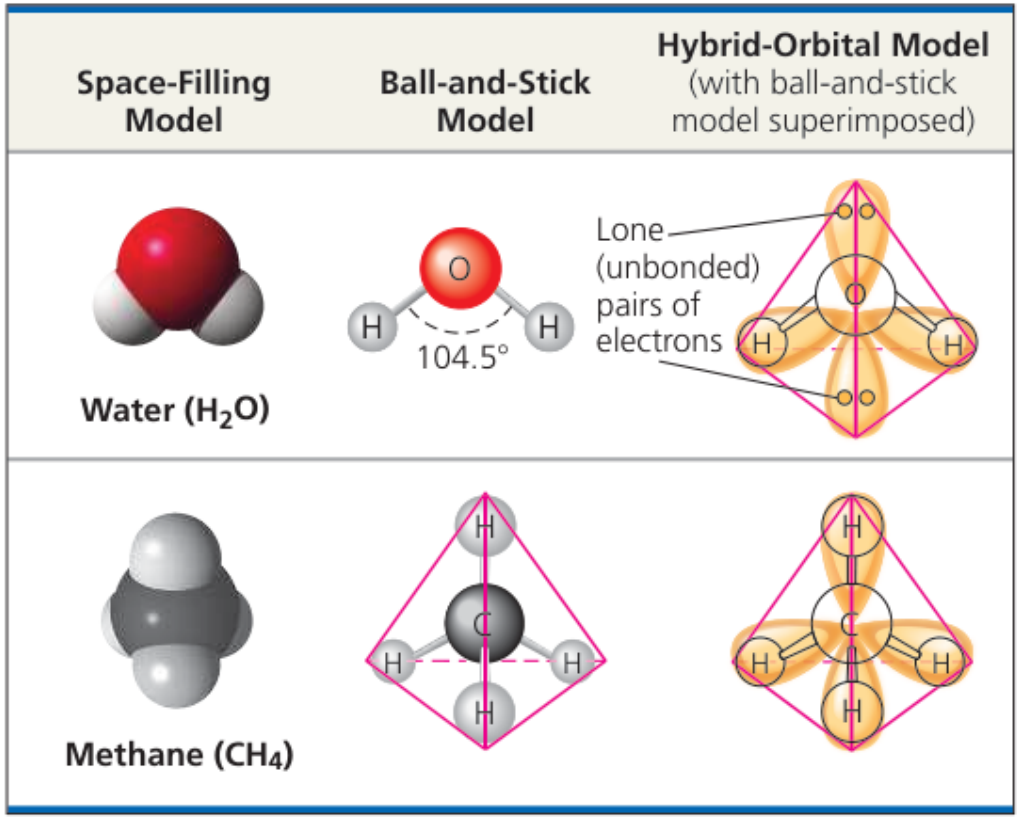
These shapes are determined by the positions of the atoms' orbitals

When an atom forms covalent bonds, the orbitals in its valence shell undergo rearrangement.

▼ Figure 2.15 Molecular shapes due to hybrid orbitals.



(a) Hybridization of orbitals. The single *s* and three *p* orbitals of a valence shell involved in covalent bonding combine to form four teardrop-shaped hybrid orbitals. These orbitals extend to the four corners of an imaginary tetrahedron (outlined in pink).

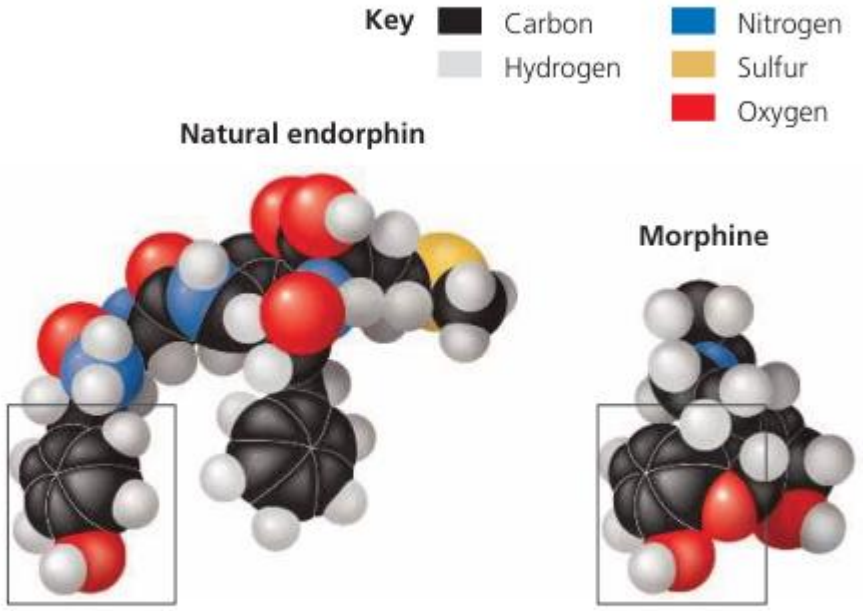


(b) Molecular-shape models. Three models representing molecular shape are shown for water and methane. The positions of the hybrid orbitals determine the shapes of the molecules.

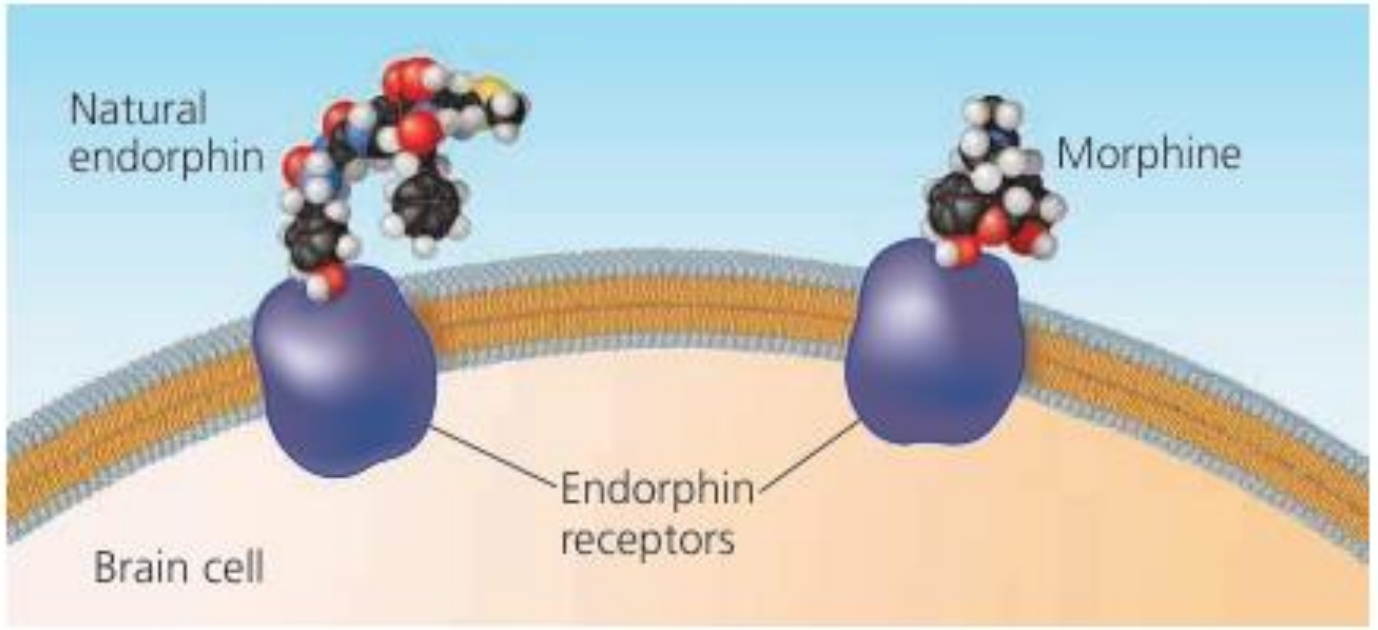
Molecular shape is key to function

Molecular shape is crucial: It determines how biological molecules recognize and respond to one another with specificity. Biological molecules often bind temporarily to each other by forming weak interactions, but only if their shapes are complementary.

▼ **Figure 2.16 A molecular mimic.** Morphine affects pain perception and emotional state by mimicking the brain's natural endorphins.



(a) Structures of endorphin and morphine. The boxed portion of the endorphin molecule (left) binds to receptor molecules on target cells in the brain. The boxed portion of the morphine molecule (right) is a close match.



(b) Binding to endorphin receptors. Both endorphin and morphine can bind to endorphin receptors on the surface of a brain cell.

Chemical reactions make and break chemical bonds

Chemical reactions lead to changes in the composition of matter

Reactants --→ Products

Matter is conserved in a chemical reaction: Reactions cannot create or destroy atoms but can only rearrange the electrons among them.

hydrogen and nitrogen molecules can combine to form ammonia, but ammonia canalso decompose to regenerate hydrogen and nitrogen:

$$3\,H_2 + N_2 \rightleftharpoons 2\,NH_3$$

The two opposite-headed arrows indicate that the reaction is reversible.

All chemical reactions are theoretically **reversible**, with the products of the forward reaction becoming the reactants for the reverse reaction.
One of the factors affecting the rate of a reaction is the concentration of reactants. The greater the concentration of reactant molecules, the more frequently they collide with one another and have an opportunity to react and form products. The same holds true for products. As products accumulate, collisions resulting in the reverse reaction become more frequent.
Chemical equilibrium is achieved when the forward and reverse reactions occur at the same rate, and the relative concentrations of products and reactants stop changing.

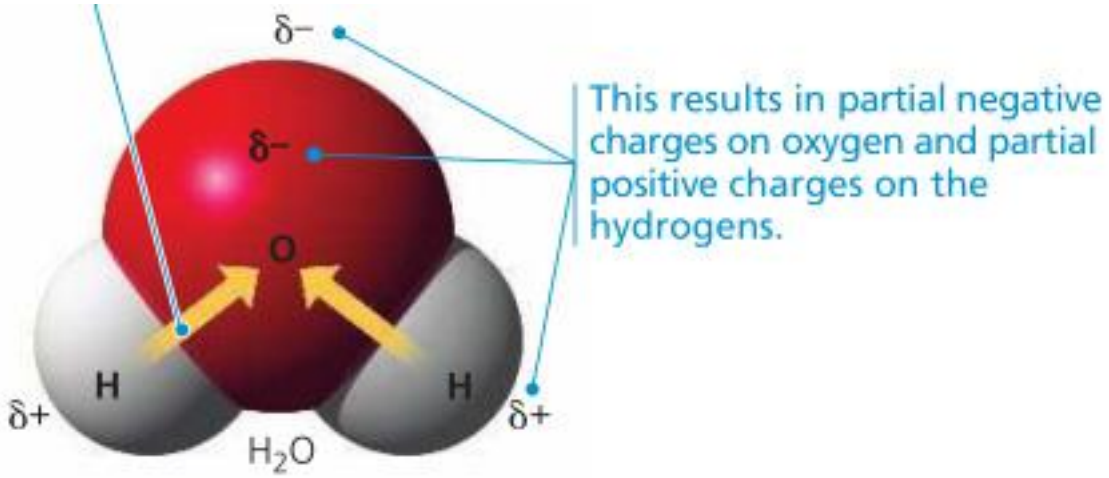
Water and Life

60% of body is water.

Regulate body temperature.
Removes wastes.
Carry nutrients and oxygen to cells.
Perform hydrolysis reactions to break food down.
Convert food to energy in cellular respiration.
Absorb nutrients.
Protect and cushion organs and joints.

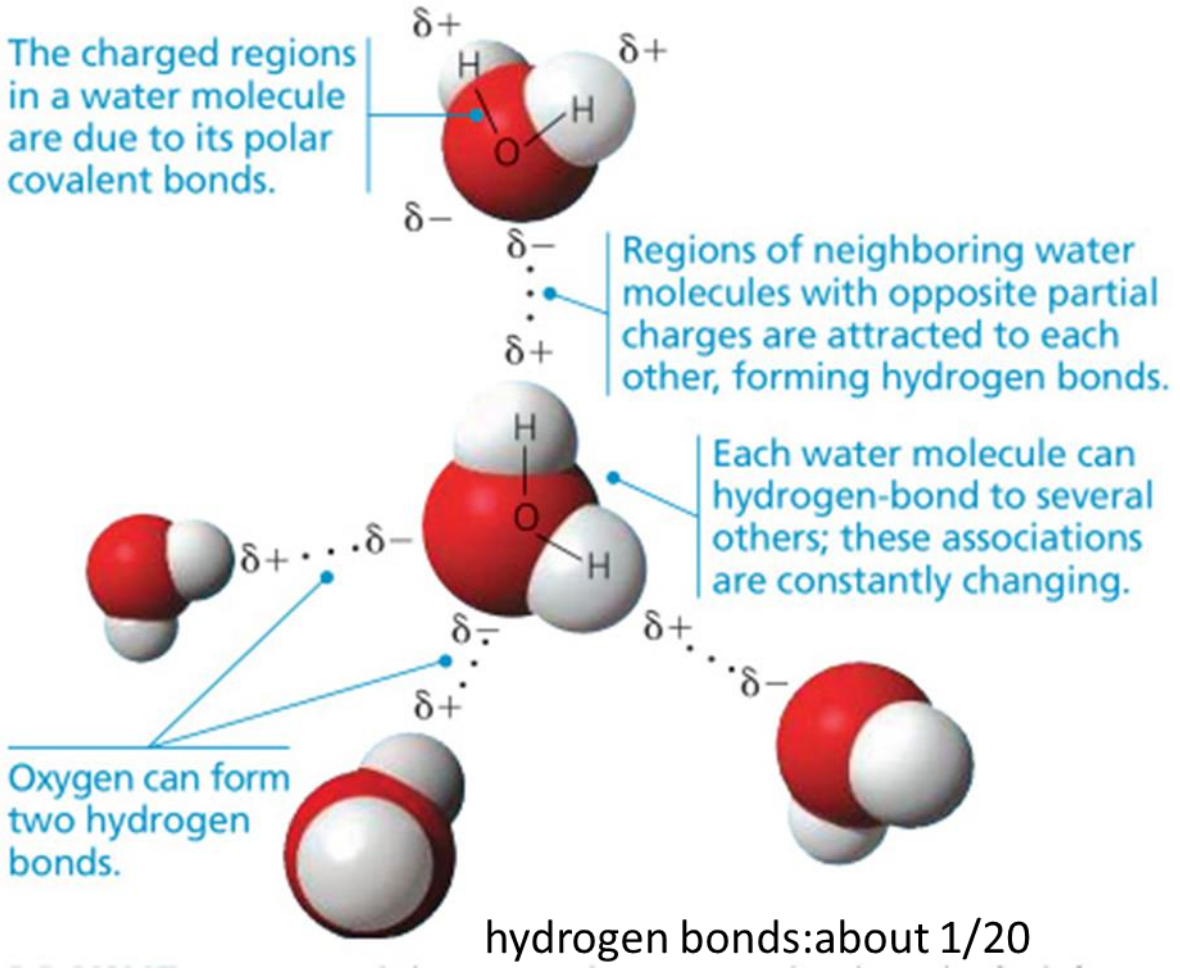
Polar covalent bonds in water molecules result in hydrogen bonding

The bonds that form water molecules are polar covalent bonds:



This unequal sharing of electrons and water's V-like shape make it a polar molecule, meaning that its overall charge is unevenly distributed. In water, the oxygen of the molecule has partial negative charges $(\delta-)$ and the hydrogens have partial positive charges $(\delta+)$

Water molecules form many hydogen bonds with neihgboring water molecules.



hydrogen bonds:about 1/20 as strong as a covalent bond.

Four properties of water which sustains life

1. Cohesion and adhesion
1. High heat capacity
2. Less dense in solid form
3. Solvent properties

Cohesion and adhesion of water molecules

Cohesion: Water molecules stay close to each other as a result of hydrogen bonding. Although the arrangement of molecules in a sample of liquid water is constantly changing, at any given moment many of the molecules are linked by multiple hydrogen bonds. Collectively, the hydrogen bonds hold the substance together, a phenomenon called cohesion. Formation of water droplets, high surface tension are due to cohesion.



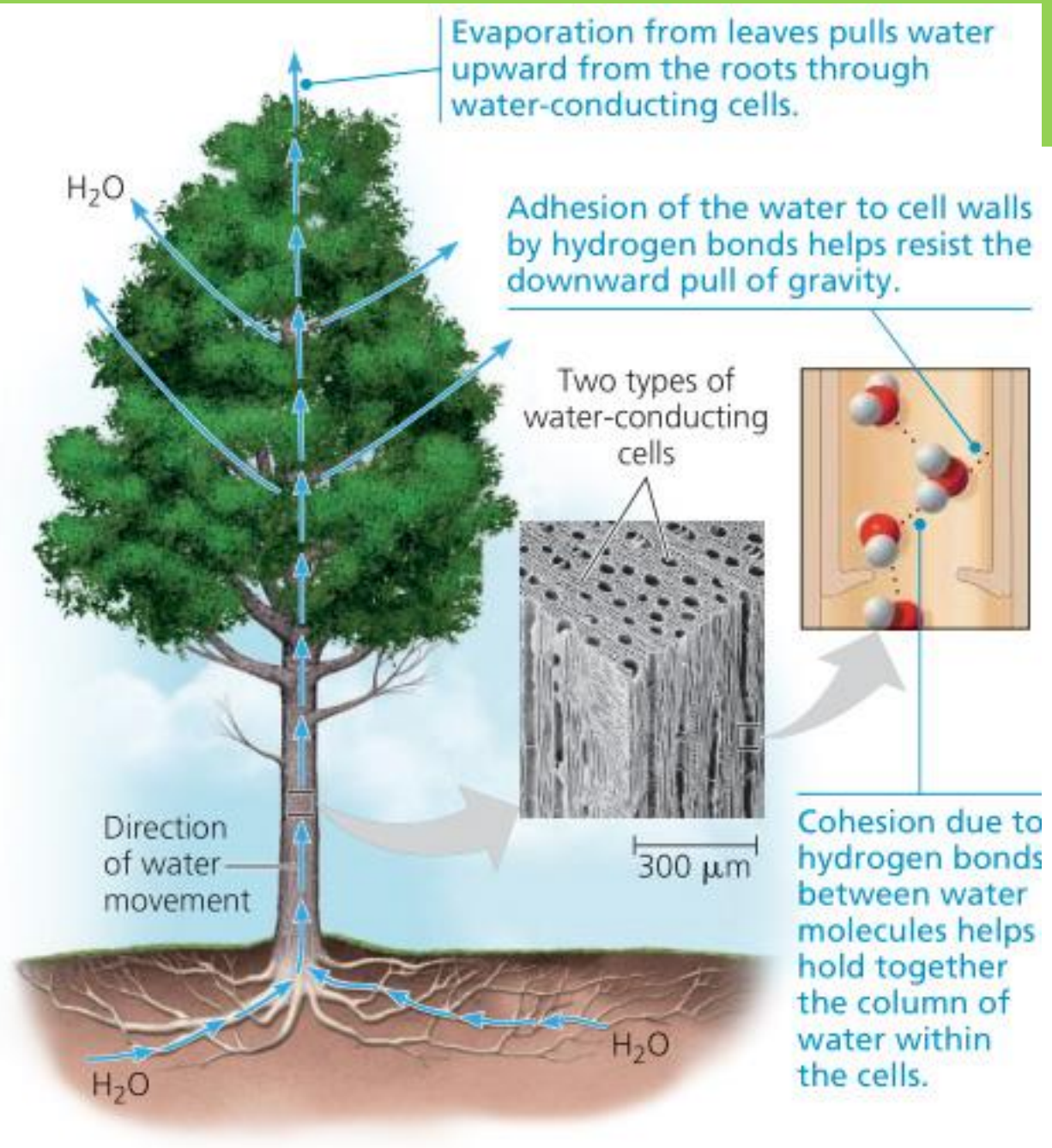


surface tension, a measure of how difficult it is to stretch or break the surface of a liquid.

Adhesion: water molecules tend to adhere to other molecules like the walls of a capillary tube.

Cohesion and adhesion of water molecules

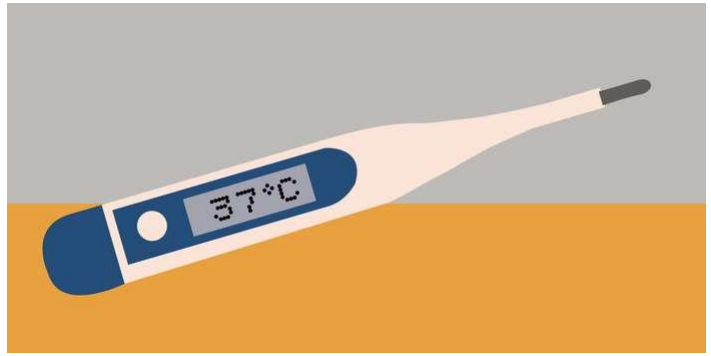
Cohesion contributes to the transport of water and dissolved nutrients against gravity in plants.
Adhesion of water by hydrogen bonds to the molecules of cell walls also helps counter the downward pull of gravity.



Moderation of temperature by water

Water moderates air temperature by absorbing heat from air that is warmer and releasing stored heat to air that is cooler.
Water has high heat capacity that is the amount of heat it takes to raise the temperature of 1 g of water by 1°C is high.
Water can absorb a relatively large amount of heat with only a slight change in its own temperature because most of the heat is used to break the hydrogen bonds between the molecules.
A calorie of heat causes a relatively small change in the temperature of water because much of the heat is used to disrupt hydrogen bonds before the water molecules can begin moving faster.
And when the temperature of water drops slightly, many additional hydrogen bonds form, releasing a considerable amount of energy in the form of heat.

Many organisms can operate within a certain range of temperatures.
Water helps maintain body temperature.

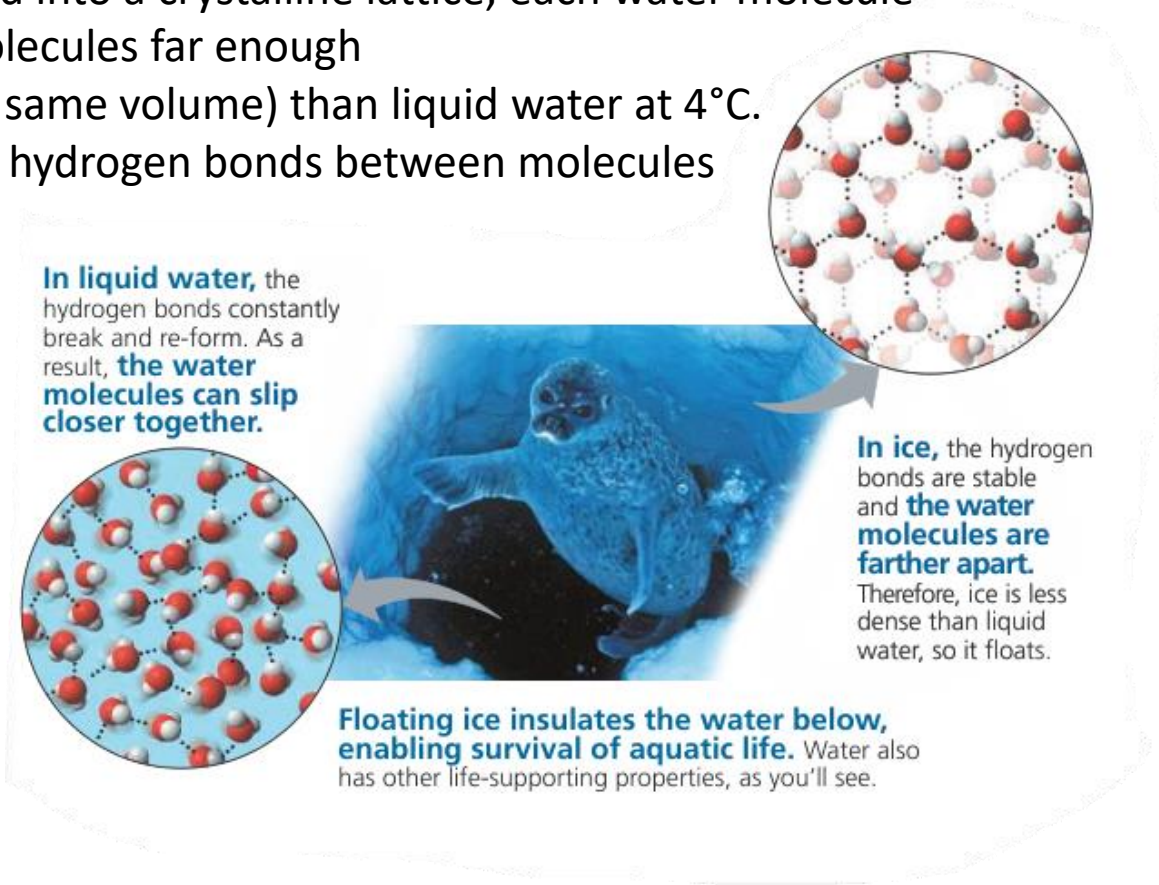


When it is hot, the water in the body absorbs heat energy and is released as sweat cooling the body down.

Water is less dense as as solid.

Water is one of the few substances that are less dense as a solid than as a liquid. As a result, ice floats on liquid water.

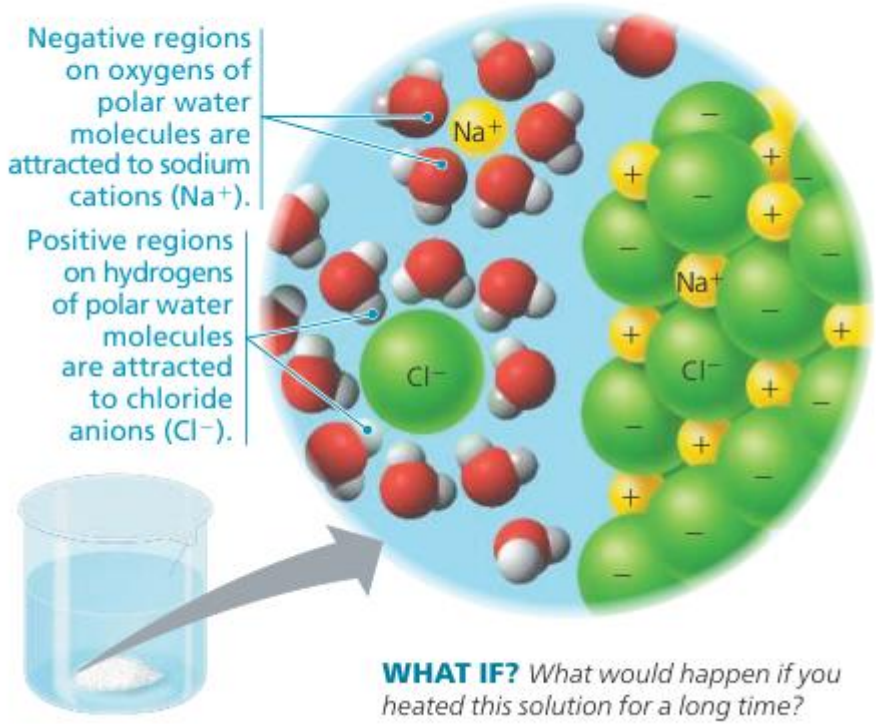
At temperatures above 4°C, water behaves like other liquids, expanding as it warms and contracting as it cools.
As the temperature falls from 4°C to 0°C, water begins to freeze because more and more of its molecules are moving too slowly to break hydrogen bonds. At 0°C, the molecules become locked into a crystalline lattice, each water molecule hydrogen-bonded to four partners. The hydrogen bonds keep the molecules far enough apart to make ice about 10% less dense (10% fewer molecules in the same volume) than liquid water at 4°C.
When ice absorbs enough heat for its temperature to rise above 0°C, hydrogen bonds between molecules are disrupted. As the crystal collapses, the ice melts and molecules have fewer hydrogen bonds, allowing them to slip closer together. Water reaches its greatest density at 4°C and then begins to expand as the molecules move faster.
The ability of ice to float due to its lower density is an important factor in the suitability of the environment for life.
when a deep body of water cools, the ice floats, insulating the liquid water below. This prevents it from freezing and allows life to exist under the frozen surface
Besides insulating the water below, ice also provides a solid habitat for some animals, such as polar bears and seals.



Solvent Properties of Water

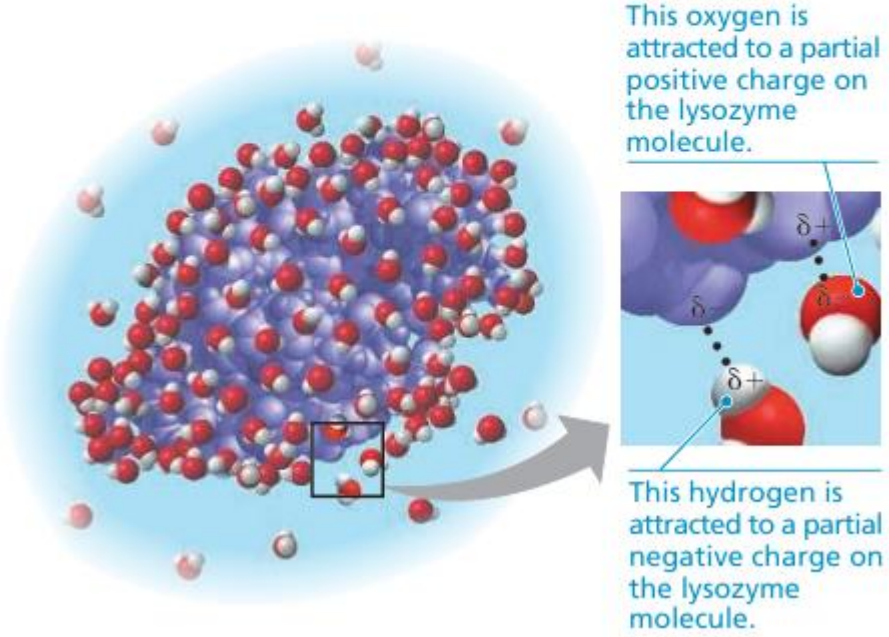
An aqueous solution is one in which the solute is dissolved in water; water is the solvent.
Water is called the universal solvent but this doesn't mean it can dissolve everything.
Can dissolve Polar molecules and ionic compounds such as salts, sugars, gases and water soluble proteins.

▼ Figure 3.8 Table salt dissolving in water. A sphere of water molecules, called a hydration shell, surrounds each solute ion.



WHAT IF? *What would happen if you heated this solution for a long time?*

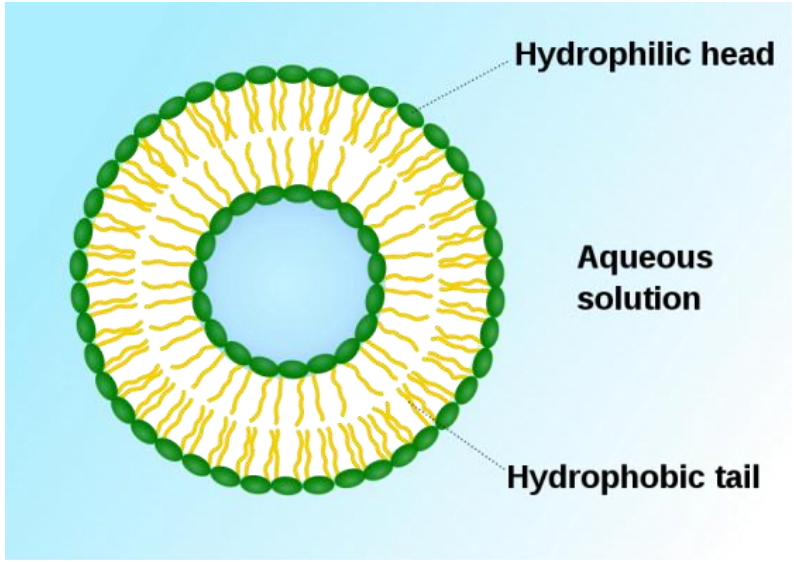
▼ Figure 3.9 A water-soluble protein. Human lysozyme is a protein found in tears and saliva that has antibacterial action (see Figure 5.16). This model shows the lysozyme molecule (purple) in an aqueous environment. Ionic and polar regions on the protein's surface attract the partially charged regions on water molecules.



Hydrophilic and Hydrophobic Substances

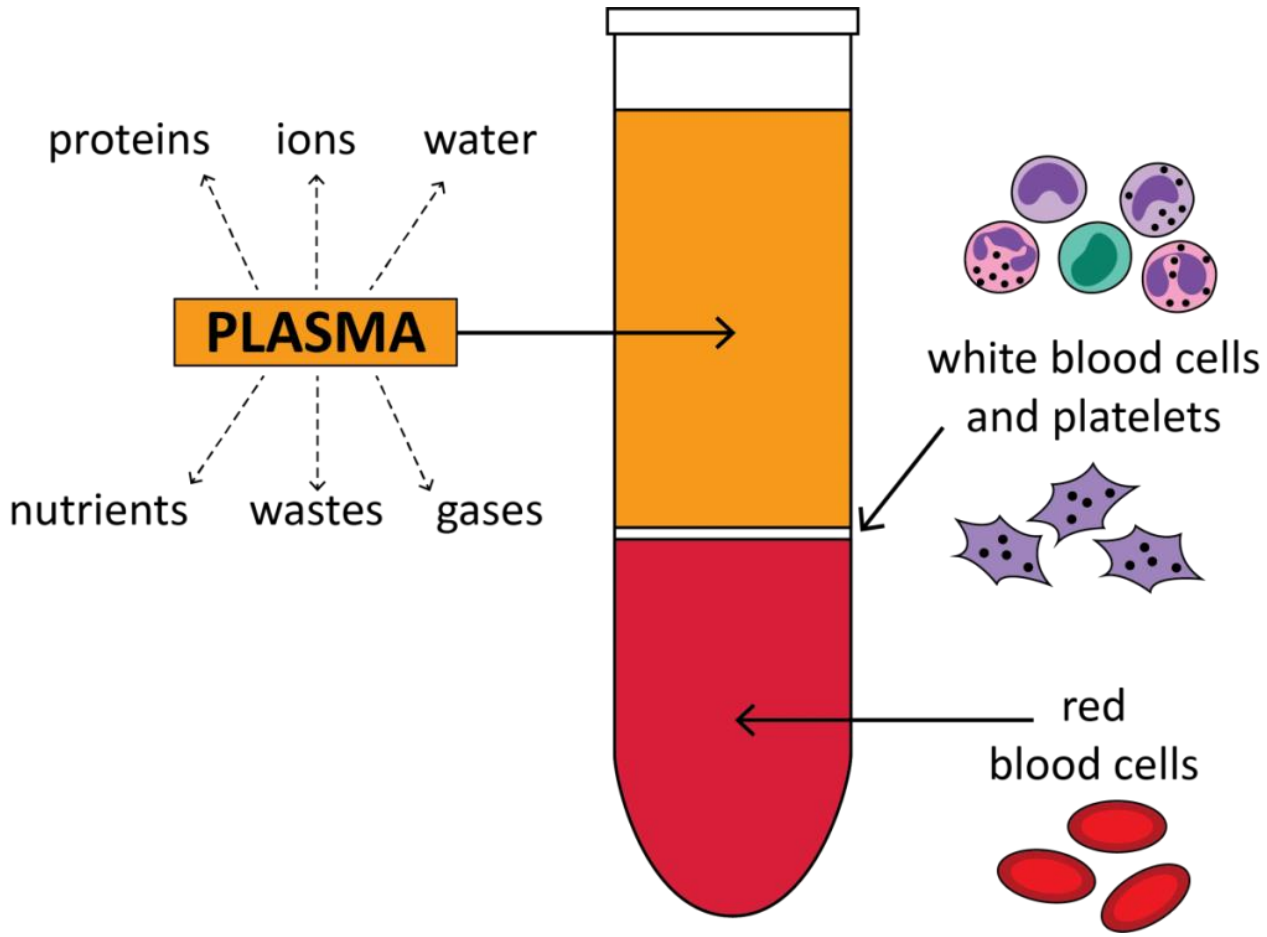
Any substance that has an affinity for water is said to be hydrophilic (from the Greek hydro, water, and philos, loving).
ex: NaCl, cotton

Substances that are nonionic and nonpolar (or otherwise cannot form hydrogen bonds) actually seem to repel water; these substances are said to be hydrophobic (from the Greek phobos, fearing).
Ex: oil,

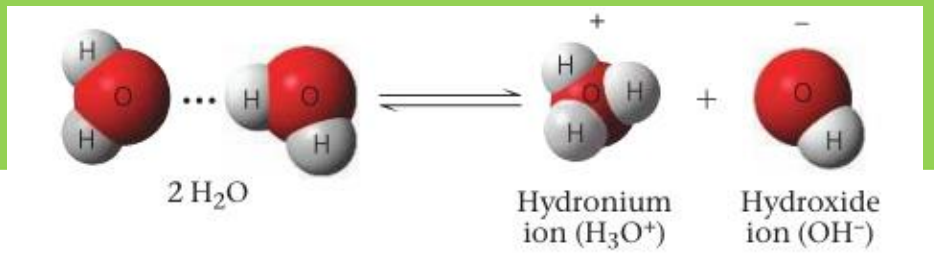


Water and Life

Water keeps undissolvable substances in suspension and allows their transportation to cells.

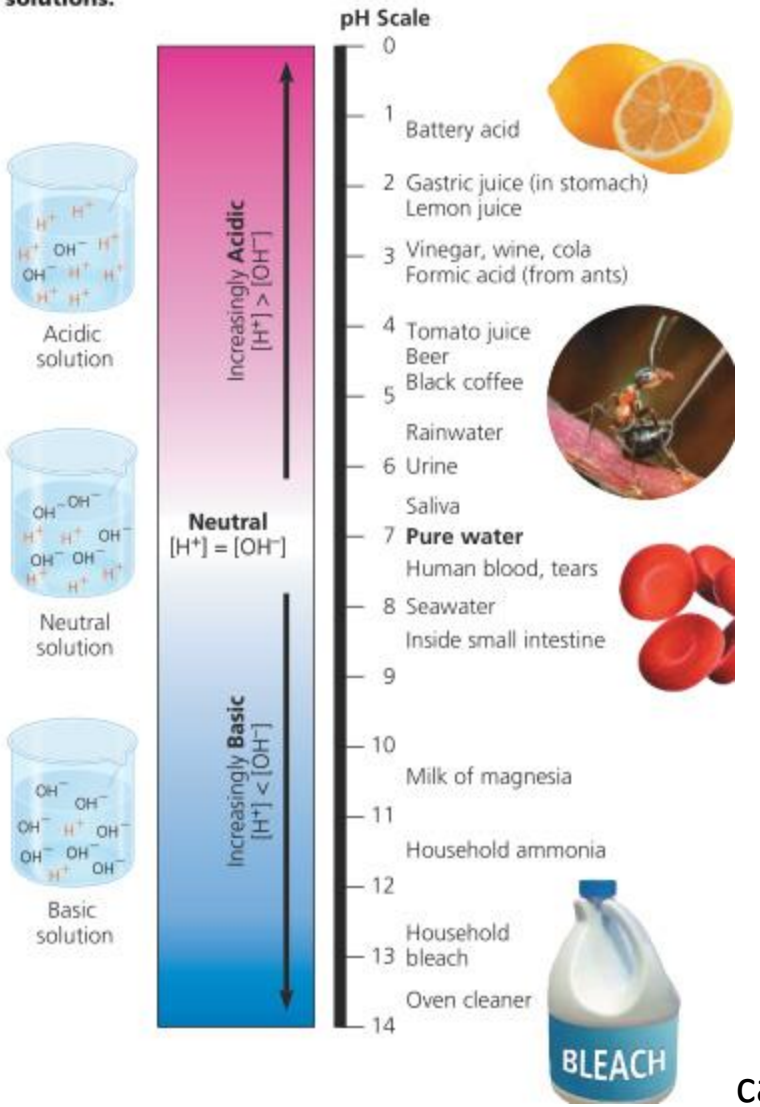


Acids, bases and pH



The pH scale (Figure 3.11) is a simple numerical method for expressing the range of H^+ concentrations. The H^+

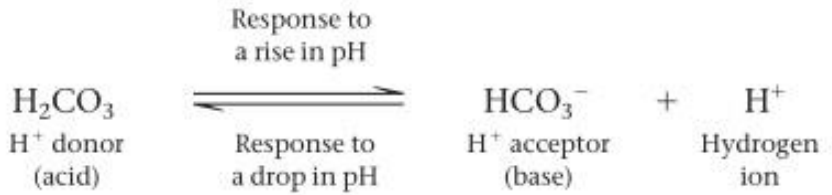
▼ **Figure 3.11 The pH scale and pH values of some aqueous solutions.**



In any aqueous solution at 25°C

$$[H^+][OH^-] = 10^{-14}$$ pH = -log [H+]

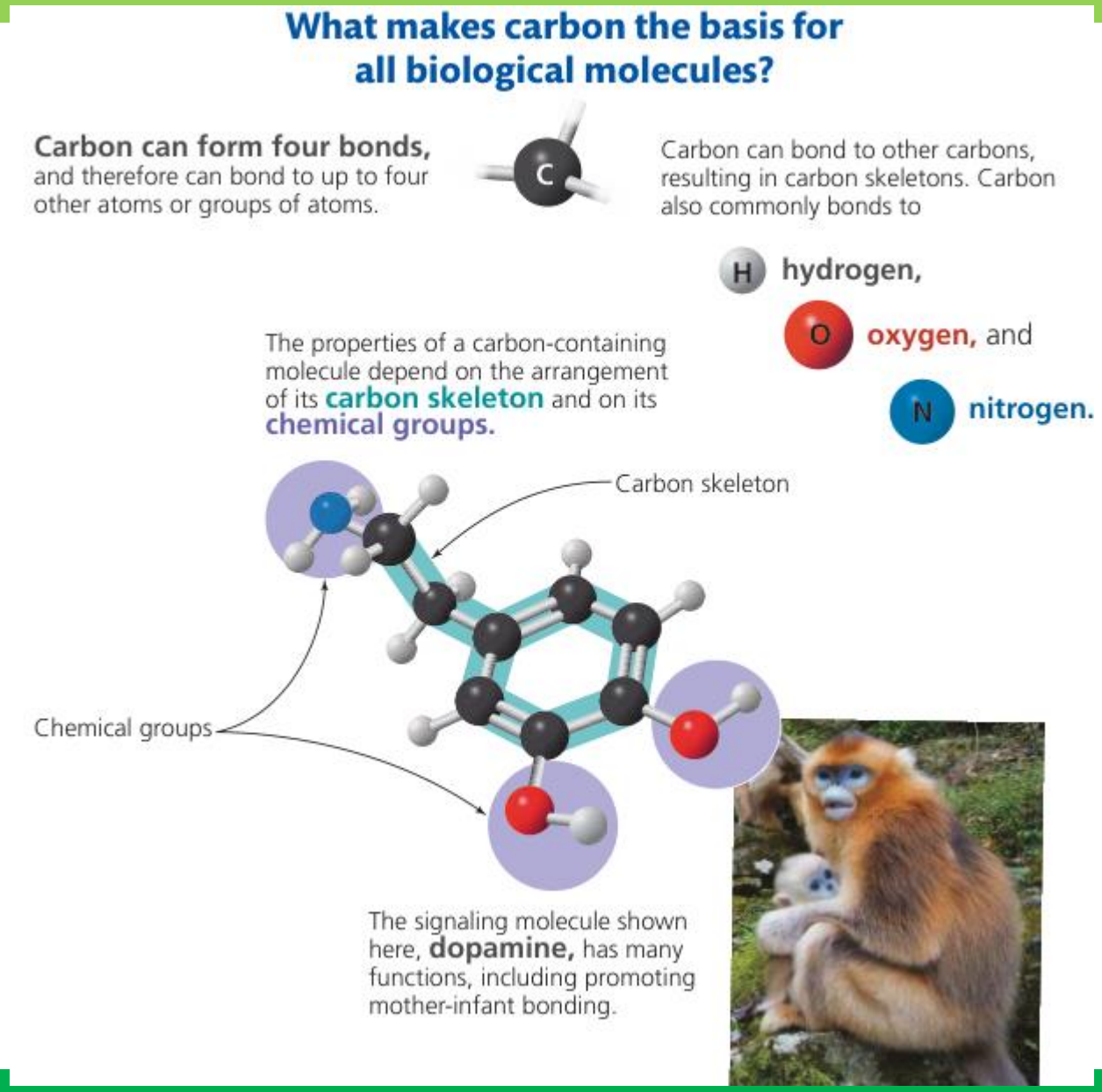
pH varies in different organs in order to allow them to carry on different functions.
Cells need to maintain pH to prevent tissue damage.
Changes in blood levels may lead to death due to acidosis or alkalosis.
Cells try to maintain pH levels by transporting different ions across the Cell membrane using proton pumps.
A buffer is a substance that minimizes changes in the concentrations of H+ and OH- in a solution. It does so by accepting hydrogen ions from the solution when they are in excess and donating hydrogen ions to the solution when they have been depleted. Most buffer solutions contain a weak acid and its corresponding base, which combine reversibly with hydrogen ions.



carbonic acid dissociates to yield a bicarbonate ion (HCO3-) and a hydrogen ion (H+)

Carbon and the Molecular Diversity of Life

- Water constitutes about 70-95% of cells. The remainder of cells is mostly made up of carbon compounds.
- Carbon can form a vast number of large, complex and diverse molecules.
- The molecules such as proteins, lipids, nucleic acids and carbohydrates that distinguish living things from non-living mater are composed of carbon.



Carbon and the Molecular Diversity of Life

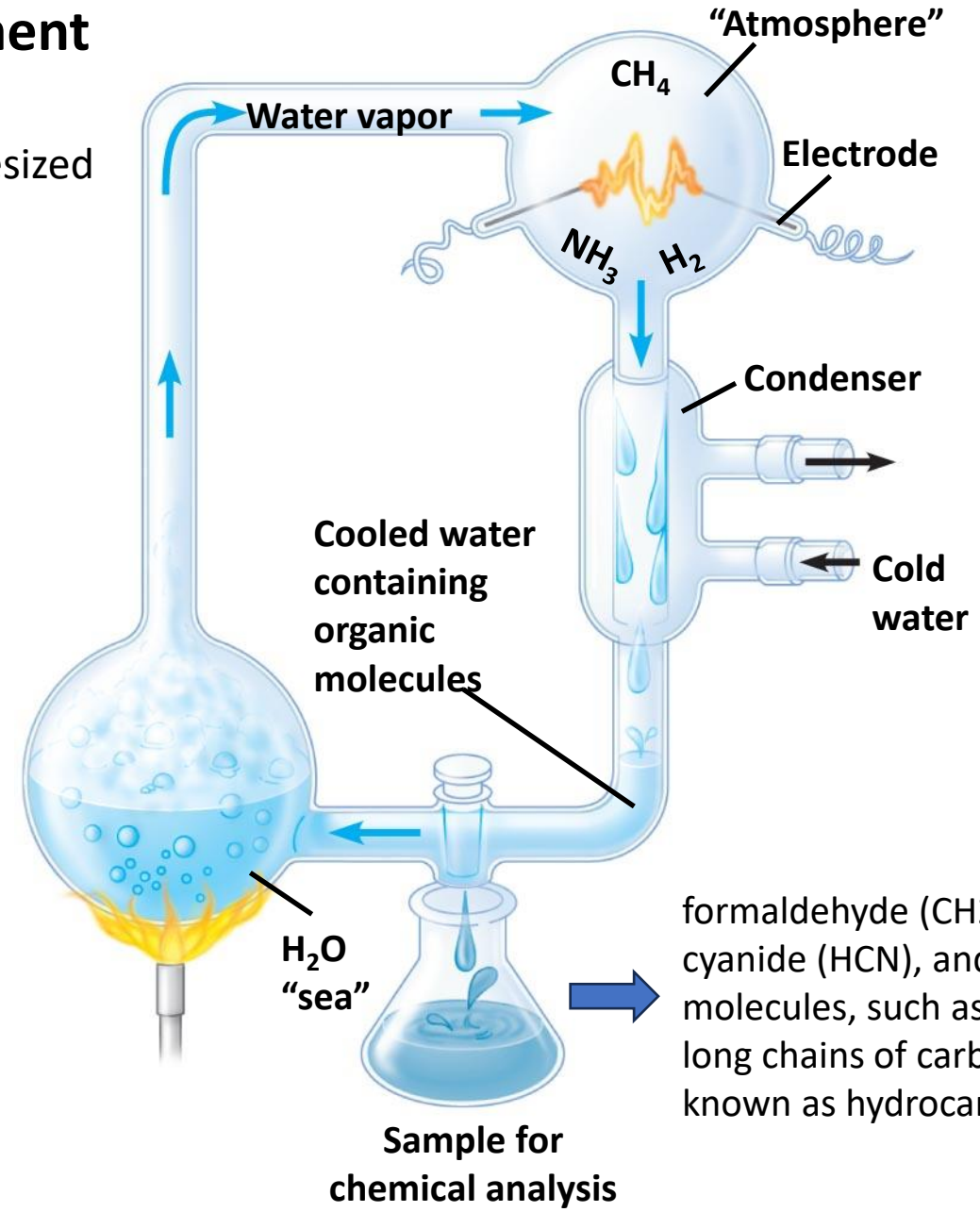
Compounds that contain carbon are called organic compounds and organic chemistry studies organic compounds.

Organic compounds can be as simple as methane or as complex as titin which is a protein made up of 27,000 amino acids.

Major elements that can be found in organic compounds in addition to carbon are H, O, N, P and S.

Stanley Miller's Experiment

1953
Could organic molecules be synthesized abiotically on the early Earth?



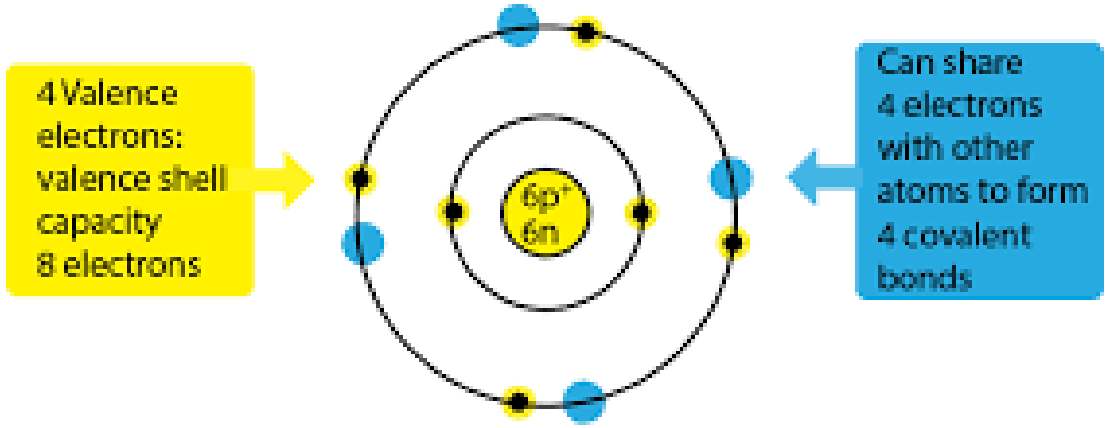
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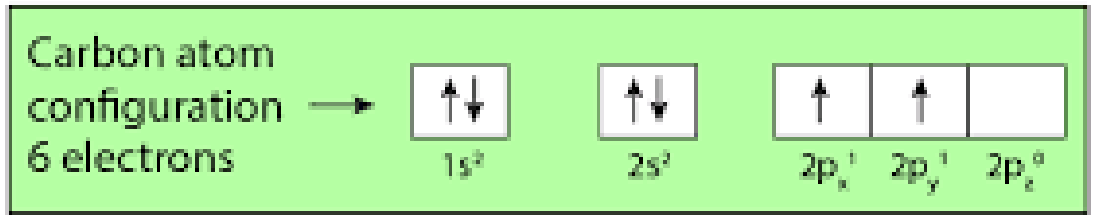
formaldehyde (CH2O) and hydrogen cyanide (HCN), and more complex molecules, such as amino acids and long chains of carbon and hydrogen known as hydrocarbon.

Carbon and the Molecular Diversity of Life

valence shell, outermost shell
valence electrons. the electrons found in the valence Shell

The number of electrons in the outermost shell of a particular atom determines its tendency to form chemical bonds with other atoms.





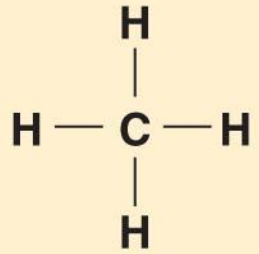
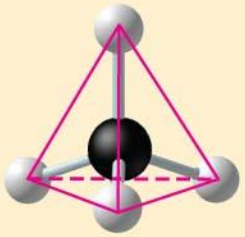
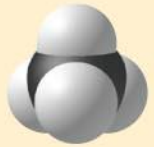
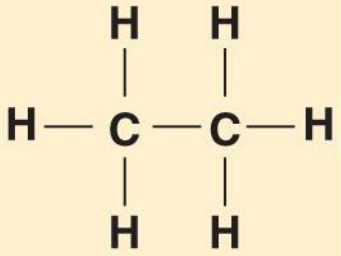
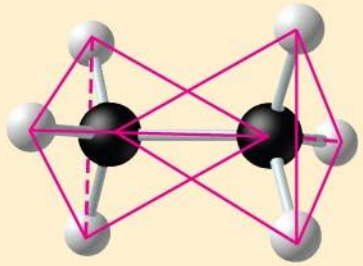
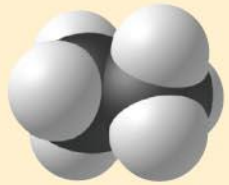
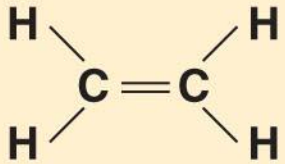
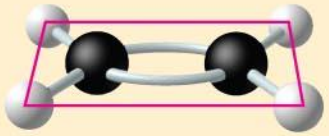
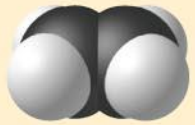
Covalent Bonds are formed when valence electrons are shared between atoms.

With four valence electrons, carbon can form 4 covalent bonds with a variety of atoms.

In organic molecules carbon usually forms single or double covalent bonds.

Carbon can act as an intersection point where as many as 4 branches arise from the atom allowing the formation of large complex molecules.

Carbon and the Molecular Diversity of Life

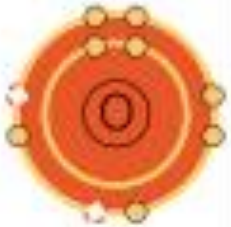
(a) Methane	CH_4			
(b) Ethane	C_2H_6			
(c) Ethene (ethylene)	C_2H_4			

all atoms
attached to those carbons
are in the same plane

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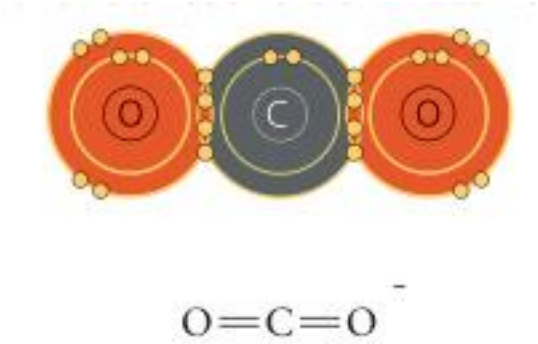
In molecules with multiple carbons, each carbon bonded to four other atoms has a tetrahedral shape However, when two carbon atoms are joined by a double bond, the molecule has a flat shape

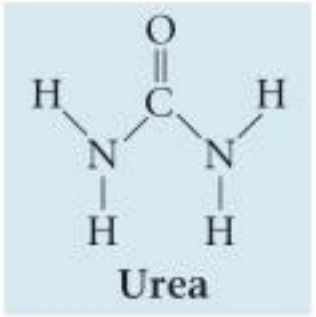
Carbon and the Molecular Diversity of Life

	Hydrogen	Oxygen	Nitrogen	Carbon
Lewis dot structure showing existing valence electrons	H·	·Ö:	·N·	·Ċ·
Electron distribution diagram with red circles showing electrons needed to fill the valence shell				
Number of electrons needed to fill the valence shell	1	2	3	4
Valence: Number of bonds the element can form	1	2	3	4

Carbon and the Molecular Diversity of Life

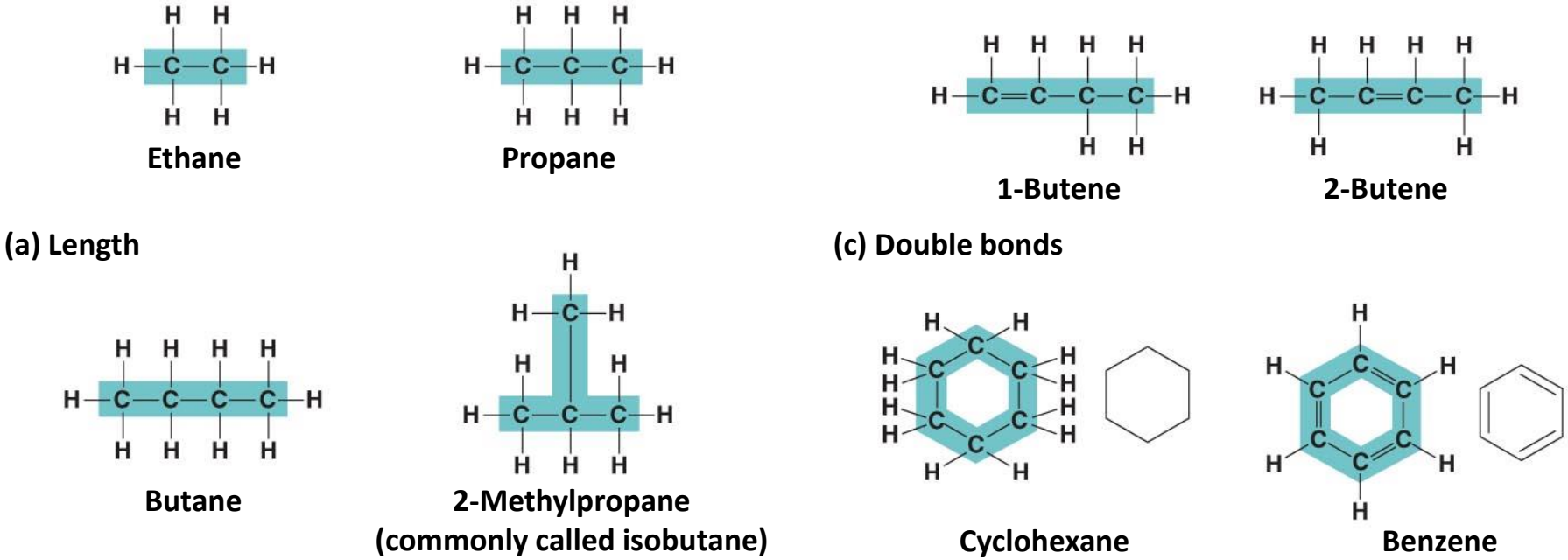
In the carbon dioxide molecule , a single carbon atom is joined to two atoms of oxygen by double covalent bonds. The structural formula for CO2 is shown here:
O=C=O
Each line in a structural formula represents a pair of shared electrons. Thus, the two double bonds in CO2 have the same number of shared electrons as four single bonds. The arrangement completes the valence shells of all atoms in the molecule:





Urea,, is an organic compound found in urine. Again, each atom has the required number of covalent bonds. In this case, one carbon atom participates in both single and double bonds.

Molecular Diversity Arising from Variation in Carbon Skeletons



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Carbon chains form the skeletons of most organic molecules.

Carbon skeletons vary in length and shape (straight, branched, ringed).

Molecular diversity arise from variations in carbon skeletons.

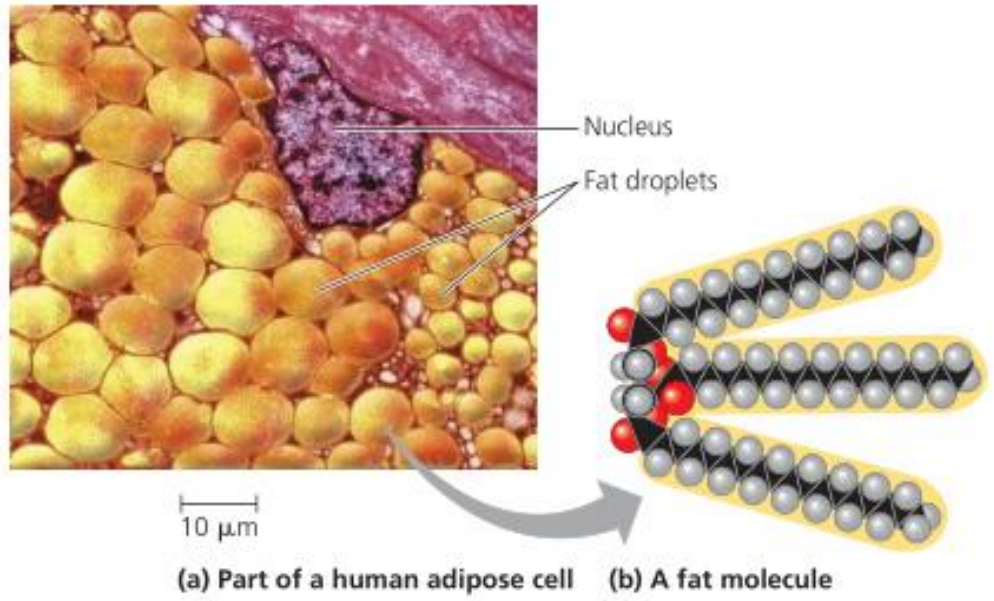
Hydrocarbons

Hydrocarbons are organic molecules consisting of only carbon and hydrogen.

Hydrocarbons can undergo reactions that release a large amount of energy. Hence gasoline is a hydrocarbon.

Many organic molecules, such as fats, have hydrocarbon components. Fats serve as stored fuel for plant seeds and animals.

Hydrocarbons are nonpolar therefore they are hydrophobic.



A fat molecule consists of a small, nonhydrocarbon component joined to three hydrocarbon tails that account for the hydrophobic behavior of fats. The tails can be broken down to provide energy.

Isomers

Isomers are compounds with the same molecular formula but different structures and properties:

Structural isomers have different covalent arrangements of their atoms

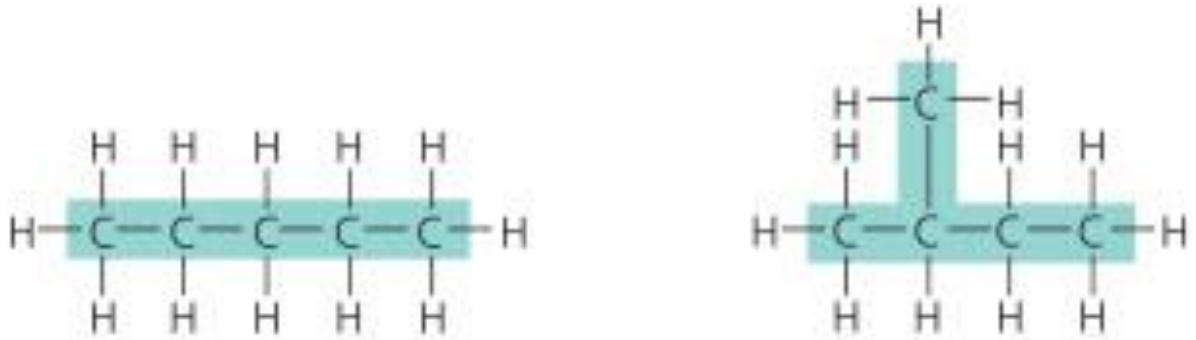
Geometric isomers have the same covalent arrangements but differ in spatial arrangements

Enantiomers are isomers that are mirror images of each other

Isomers

Structural isomers have different covalent arrangements of their atoms

(a) Structural isomers



Pentane

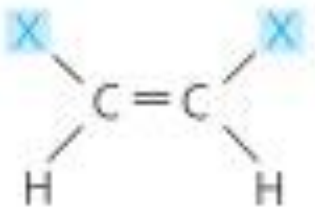
2-Methylbutane

Structural isomers differ in the arrangement of covalent bonding partners, as shown in these two isomers of C_5H_{12}.

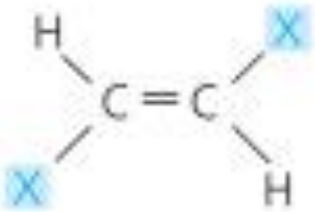
Isomers

Single bonds allow the atoms they join to rotate freely about the bond axis without changing the compound. In contrast, double bonds do not permit such rotation. **Geometric isomers** have the same covalent arrangements but differ in spatial arrangements.

(b) *Cis-trans* isomers (also known as geometric isomers)



cis isomer: The two Xs are on the same side.

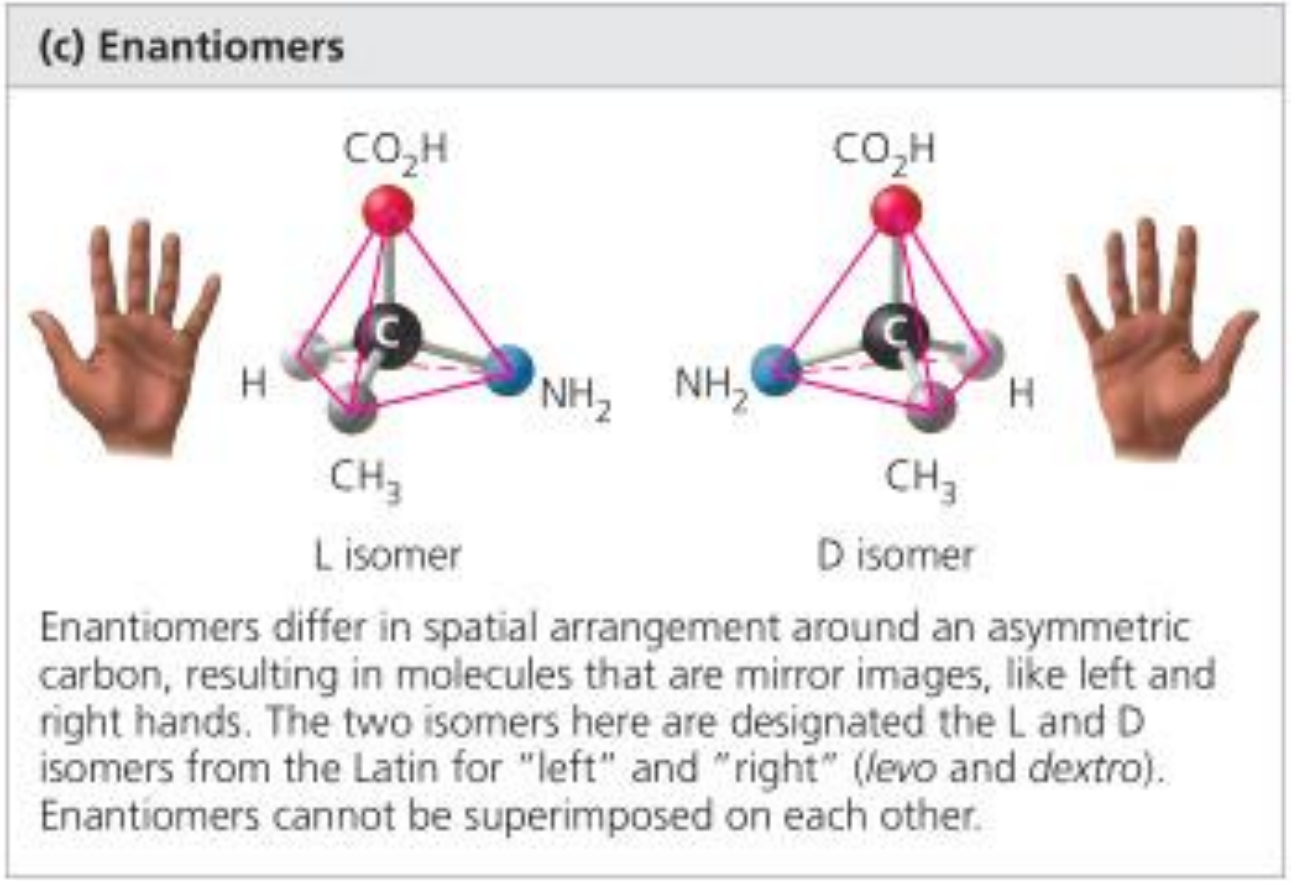


trans isomer: The two Xs are on opposite sides.

Cis-trans isomers differ in arrangement about a double bond. In these diagrams, X represents an atom or group of atoms attached to a double-bonded carbon.

Isomers

Enantiomers are isomers that are mirror images of each other.
Usually, only one isomer is biologically active because only that form can bind to specific molecules in an organism.

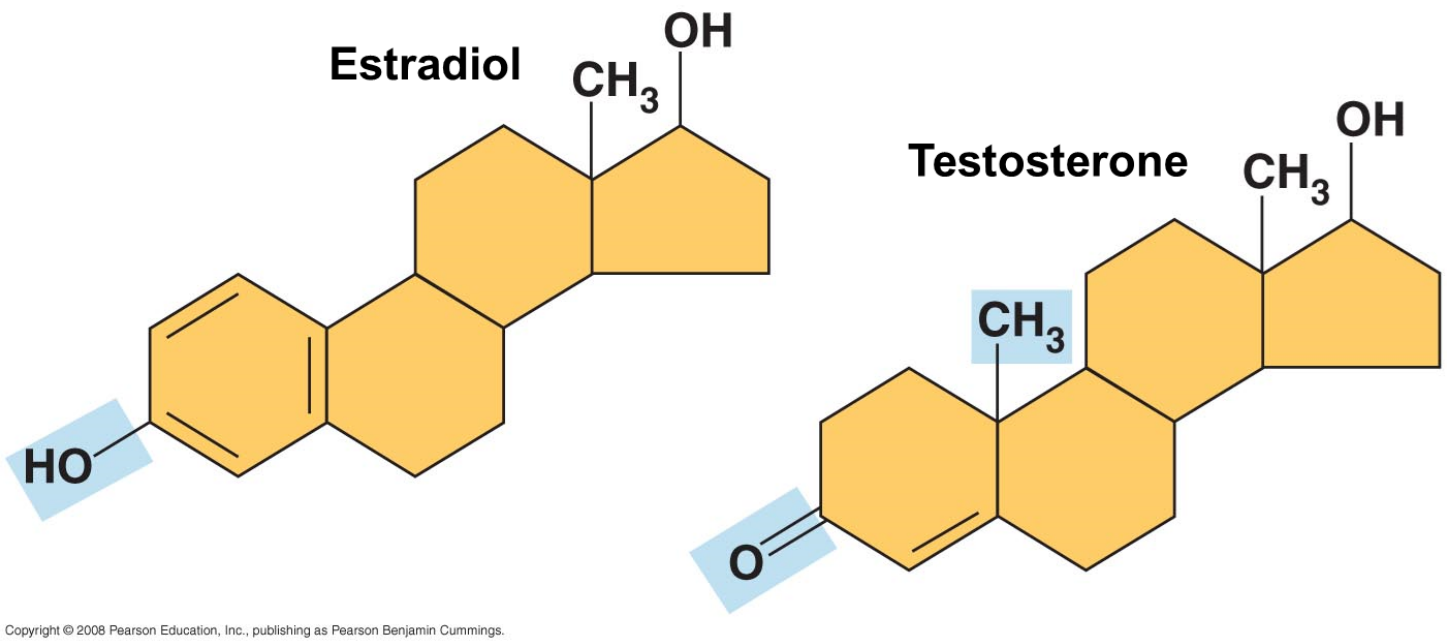


Enantiomers are important in the pharmaceutical industry
Two enantiomers of a drug may have different effects (ibuprofen, albuterol, methamphetamine)
Differing effects of enantiomers demonstrate that organisms are sensitive to even subtle variations in molecules.

A small number of chemical groups are key to the functioning of biological molecules

Distinctive properties of organic molecules depend not only on the carbon skeleton but also on the chemical groups attached to it.

These groups may participate in chemical reactions or may contribute to function indirectly by their effects on molecular shape; they help give each molecule its unique properties.



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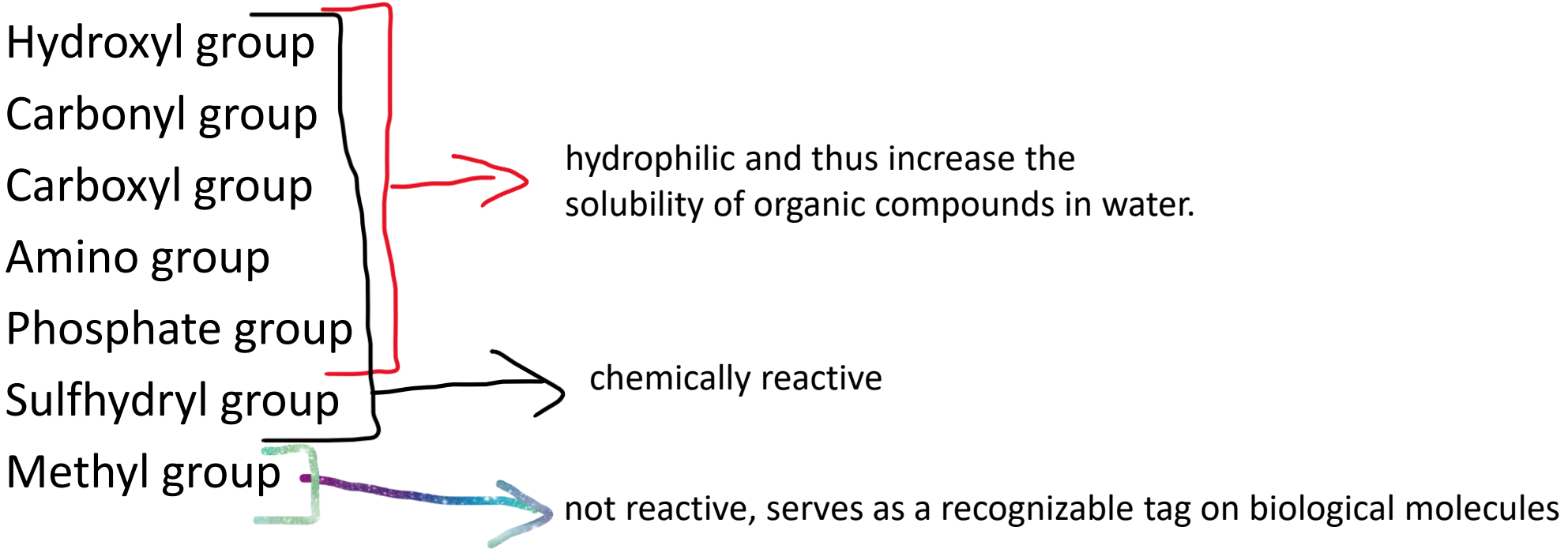
The different actions of these two molecules on many targets throughout the body are the basis of sexual characteristics, producing the contrasting features of male and female vertebrates.
In this case, the chemical groups are important because they affect molecular shape, contributing to function.

A small number of chemical groups are key to the functioning of biological molecules

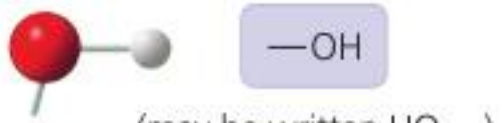
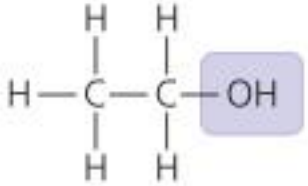
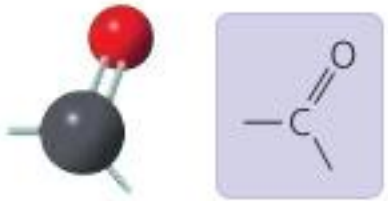
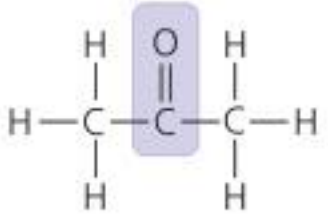
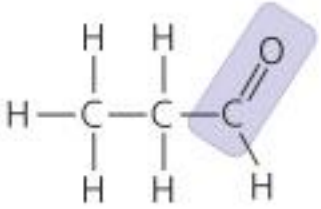
Functional groups are the components of organic molecules that are most commonly involved in chemical reactions. Each has certain properties, such as shape and charge, that cause it to participate in chemical reactions in a characteristic way.

The number and arrangement of functional groups give each molecule its unique properties.

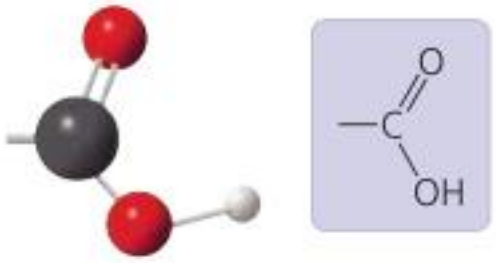
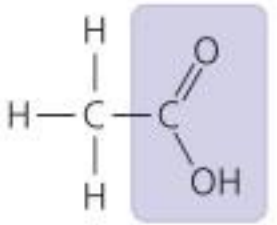
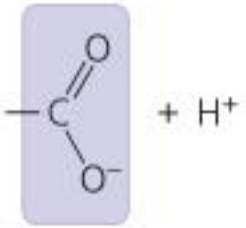
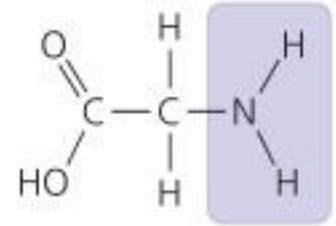
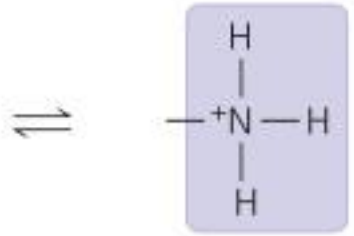
The seven functional groups that are most important in the chemistry of life:

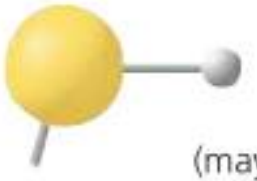
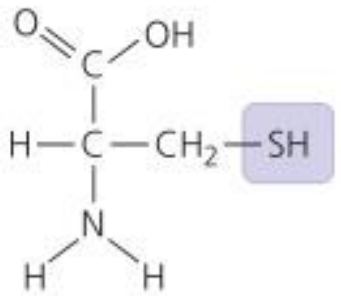
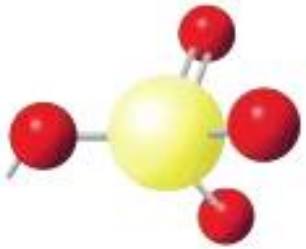
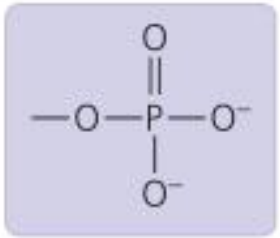
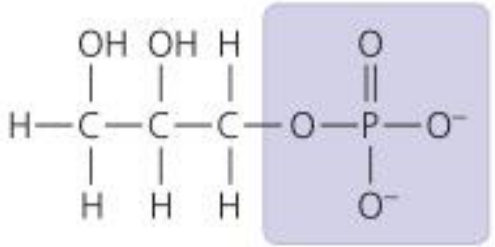
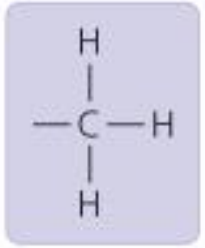
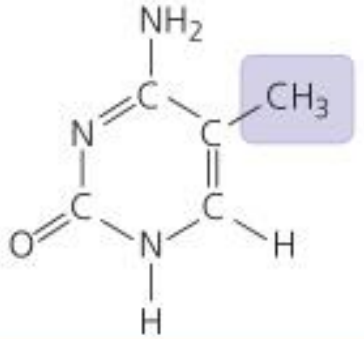


Some biologically important chemical groups.

Chemical Group	Group Properties and Compound Name	Examples
Hydroxyl group (—OH)  (may be written HO—)	Is polar due to electronegative oxygen. Forms hydrogen bonds with water, helping dissolve compounds such as sugars. Compound name: **Alcohol** (specific name usually ends in *-ol*)	 **Ethanol**, the alcohol present in alcoholic beverages
Carbonyl group (>C=O)  	Sugars with ketone groups are called ketoses; those with aldehydes are called aldoses. Compound name: **Ketone** (carbonyl group is within a carbon skeleton) or **aldehyde** (carbonyl group is at the end of a carbon skeleton)	 **Acetone**, the simplest ketone  **Propanal**, an aldehyde

Some biologically important chemical groups.

Chemical Group	Group Properties and Compound Name	Examples
Carboxyl group (—COOH) 	Acts as an acid (can donate H^+) because the covalent bond between oxygen and hydrogen is so polar. Compound name: **Carboxylic acid**, or **organic acid**	 **Acetic acid**, which gives vinegar its sour taste  Ionized form of —COOH (carboxylate ion), found in cells
Amino group ($—NH_2$) 	Acts as a base; can pick up an H^+ from the surrounding solution (water, in living organisms). Compound name: **Amine**	 **Glycine**, an amino acid (note its carboxyl group)  Ionized form of $—NH_2$, found in cells

Chemical Group	Group Properties and Compound Name	Examples
Sulfhydryl group (—SH)  —SH (may be written HS —)	Two —SH groups can react, forming a "cross-link" that helps stabilize protein structure. Hair protein cross-links maintain the straightness or curliness of hair; in hair salons, "permanent" treatments break cross-links, then re-form them while the hair is in the desired shape. Compound name: **Thiol**	 **Cysteine**, a sulfur-containing amino acid
Phosphate group ($—OPO_3^{2-}$)  	Contributes negative charge (1– when positioned inside a chain of phosphates; 2– when at the end). When attached, confers on a molecule the ability to react with water, releasing energy. Compound name: **Organic phosphate**	 **Glycerol phosphate**, which takes part in many important chemical reactions in cells
Methyl group ($—CH_3$)  	Affects the expression of genes when bonded to DNA or to proteins that bind to DNA. Affects the shape and function of male and female sex hormones. Compound name: **Methylated compound**	 **5-Methylcytosine**: Cytosine, a component of DNA, has been modified by addition of a methyl group.

ATP: An Important Source of Energy for Cellular Processes

ATP (adenosine triphosphate) is the primary energy transferring molecule in the cell.
ATP consists of an organic molecule called adenosine attached to a string of three phosphate groups.
ATP stores potential energy in its phosphate bonds.

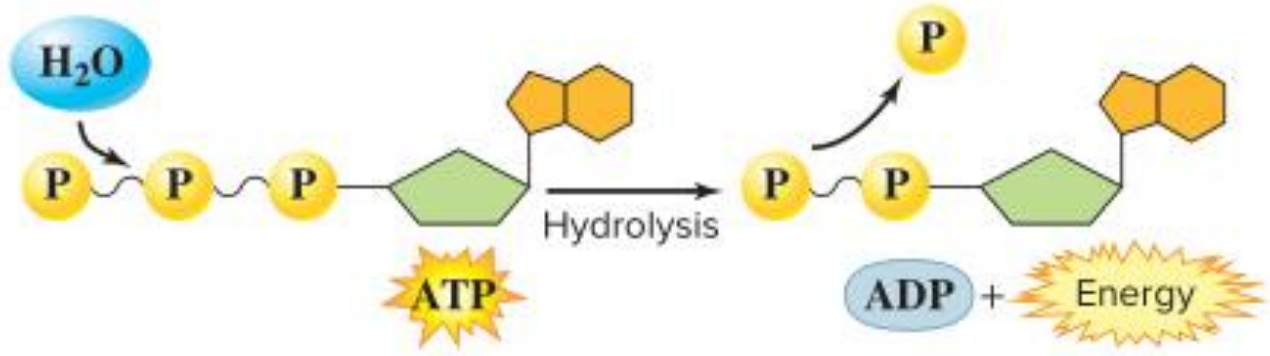


Figure 4.8 ATP Hydrolysis. Removing the endmost phosphate group of ATP yields ADP and a free phosphate group. The cell uses the released energy to do work.

The Structure and Function of Large Biological Molecules

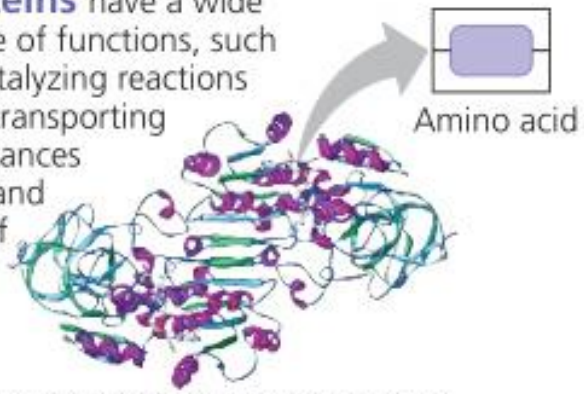
- All living things are made up of four classes of large biological molecules:
  - Carbohydrates
  - lipids
  - proteins
  - nucleic acids

Carbohydrates are a source of energy and provide structural support.



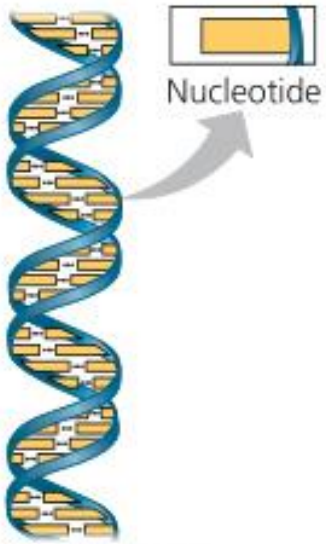
Carbohydrate (starch)

Proteins have a wide range of functions, such as catalyzing reactions and transporting substances into and out of cells.



Protein (alcohol dehydrogenase)

Nucleic acids store genetic information and function in gene expression.



Nucleic acid (DNA)

The fourth class, lipids, are not polymers or macromolecules.

Lipids are a group of diverse molecules that do not mix well with water. Key functions include providing energy, making up cell membranes, and acting as hormones.



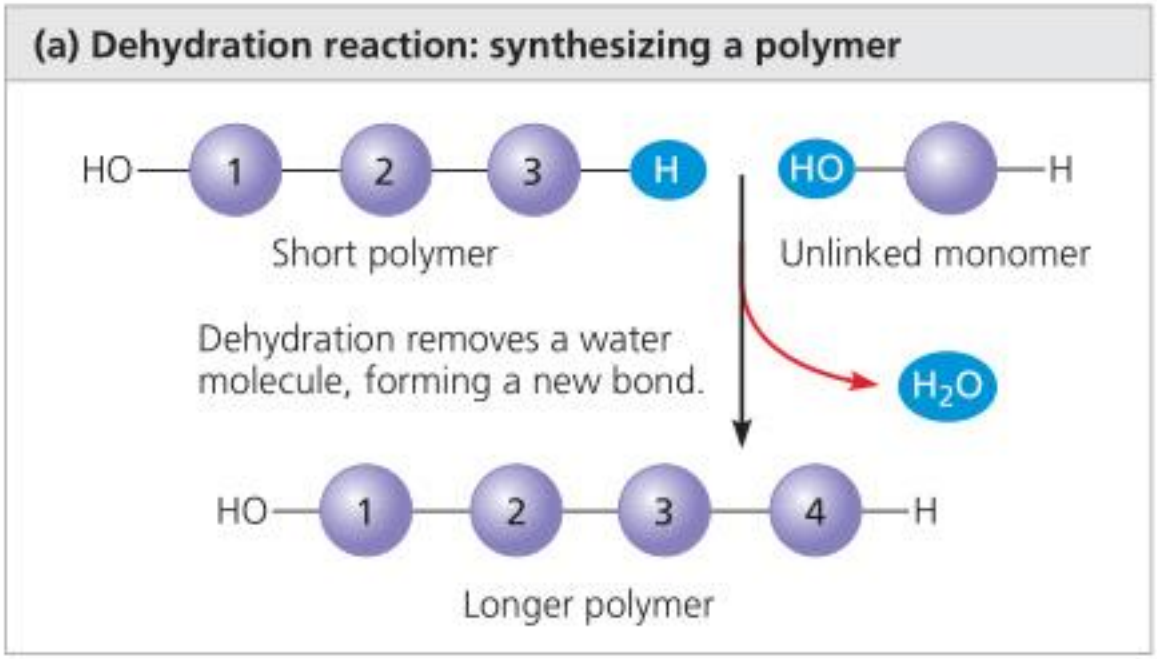
Lipid (phospholipid)

Macromolecules are polymers built from monomers

- Macromolecules are large molecules that are made up of thousands of covalently connected atoms.
- Structure of macromolecules determines function.
- Macromolecules are polymers.
- Three of the four classes of life's organic molecules –carbohydrates, proteins and nucleic acids- are polymers.
- A polymer is a long chain-like molecule consisting of many similar building blocks.
- These small building-block molecules are called monomers.

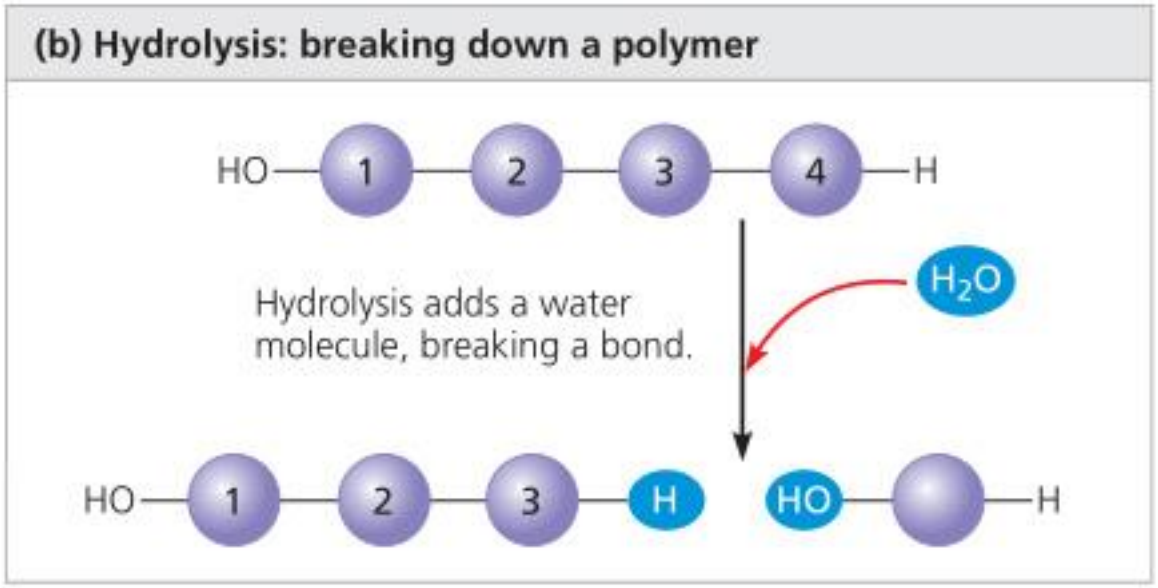
The Synthesis and Breakdown of Polymers

Synthesis of polymers: 2 monomers covalently bond together through the loss of a water molecule.
This is called a dehydration (condensation) reaction.



The Synthesis and Breakdown of Polymers

Breakdown of polymers: Polymers are disassembled to monomers by **hydrolysis**, a reaction that is essentially the reverse of the dehydration reaction.



Within the cells dehydration and hydrolysis are catalyzed by enzymes, which are macromolecules (proteins or RNA) that speed up chemical reactions.

The Diversity of Polymers

Each cell has thousands of different macromolecules .

• Macromolecules vary among cells of an organism, vary more within a species, and vary even more between species.

• An immense variety of polymers can be built from a small set of 40-50 monomers.

Variety arises from different arrangements of monomers.

Carbohydrates serve as fuel and building material

Carbohydrates consists of sugars and the polymers of sugars

- The simplest carbohydrates are monosaccharides, or single sugars
- Carbohydrate macromolecules are polysaccharides, polymers composed of many sugar building blocks

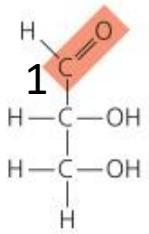
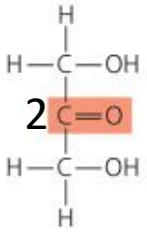
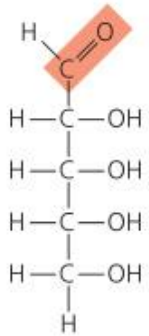
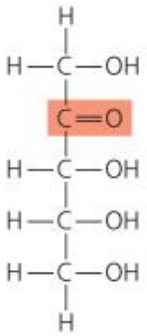
Monosaccharides = Sugars

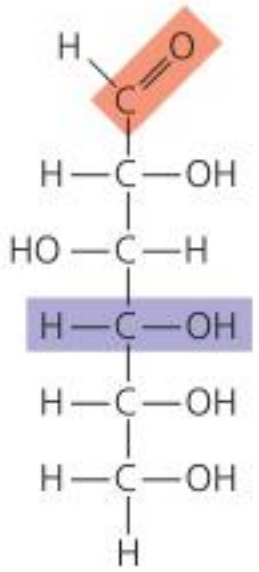
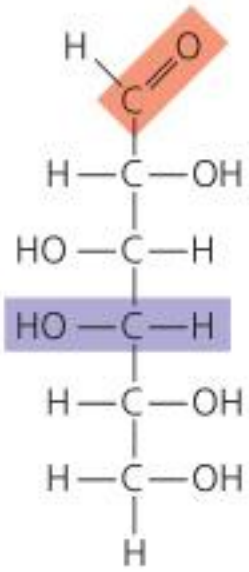
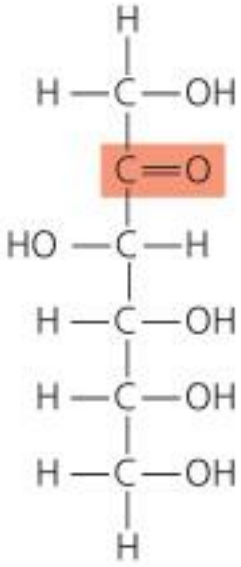
Monosaccharides have molecular formulas that are usually multiples of **CH_2O**

- Glucose ($C_6H_{12}O_6$) is the most common monosaccharide

- Monosaccharides are classified by
  - The location of the carbonyl group (as aldose or ketose)
  - The number of carbons in the carbon skeleton

Monosaccharides = Sugars

Aldoses (Aldehyde Sugars) Carbonyl group at end of carbon skeleton	Ketoses (Ketone Sugars) Carbonyl group within carbon skeleton
Trioses: three-carbon sugars ($C_3H_6O_3$)	
 **Glyceraldehyde** An initial breakdown product of glucose	 **Dihydroxyacetone** An initial breakdown product of glucose
Pentoses: five-carbon sugars ($C_5H_{10}O_5$)	
 **Ribose** A component of RNA	 **Ribulose** An intermediate in photosynthesis

Aldoses (Aldehyde Sugars) Carbonyl group at end of carbon skeleton	Ketoses (Ketone Sugars) Carbonyl group within carbon skeleton
Hexoses: six-carbon sugars ($C_6H_{12}O_6$)	
 **Glucose**  **Galactose** Energy sources for organisms	 **Fructose** An energy source for organisms

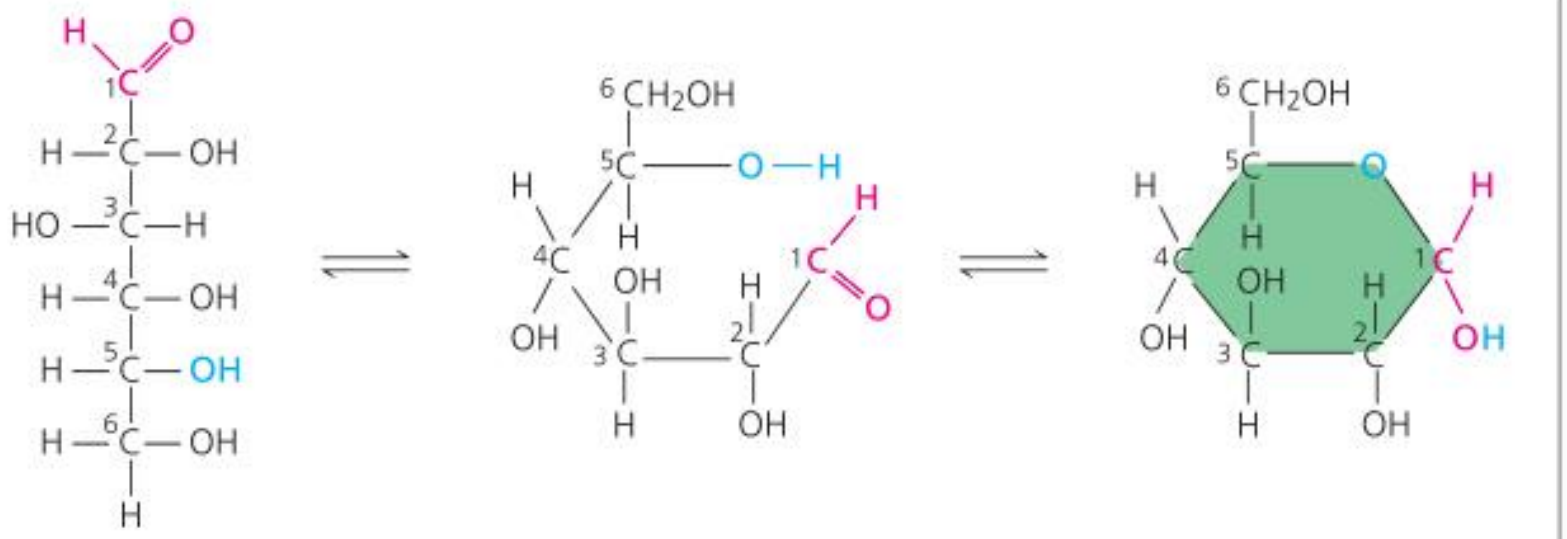
What type of isomers are glucose and galactose?

What type of isomers are glucose and fructose?

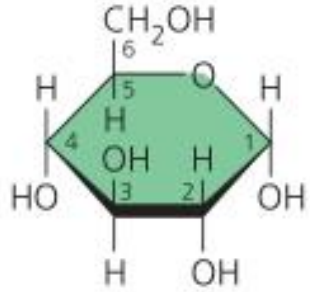
Monosaccharides = Sugars

Though often drawn as linear skeletons, in aqueous solutions many sugars form rings. because they are the most stable form of these sugars under physiological conditions

▼ **Figure 5.4 Linear and ring forms of glucose.**



(a) Linear and ring forms. Chemical equilibrium between the linear and ring structures greatly favors the formation of rings. The carbons of the sugar are numbered 1 to 6, as shown. To form the glucose ring, carbon 1 (magenta) bonds to the oxygen (blue) attached to carbon 5.



(b) Abbreviated ring structure. Each unlabeled corner represents a carbon. The ring's thicker edge indicates that you are looking at the ring edge-on; the components attached to the ring lie above or below the plane of the ring.

Monosaccharides = Sugars

Monosaccharides, particularly glucose, are major nutrients (fuel) for cells.

In the process known as cellular respiration, cells extract energy from glucose molecules by breaking them down in a series of reactions.

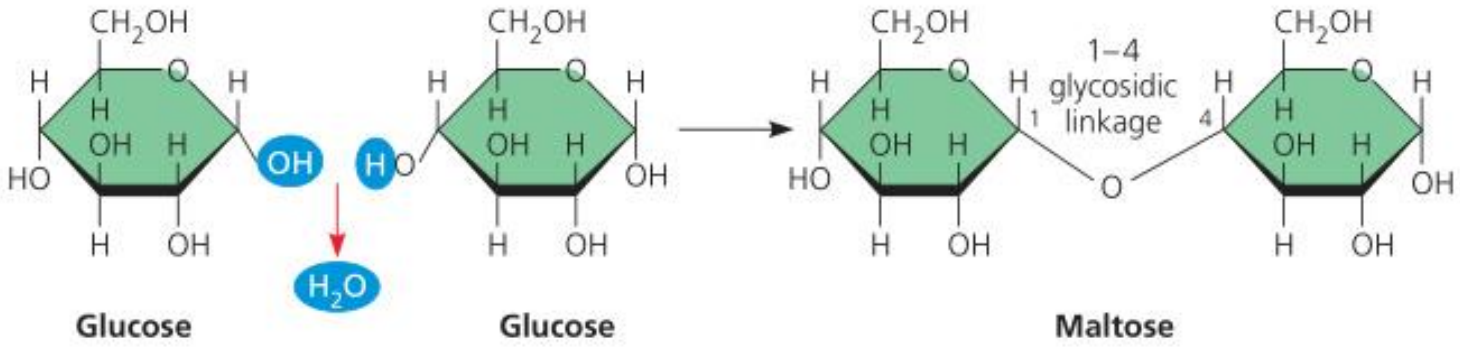
Their carbon skeletons also serve as raw material for the synthesis of other types of small organic molecules, such as amino acids and fatty acids.

Disaccharide synthesis

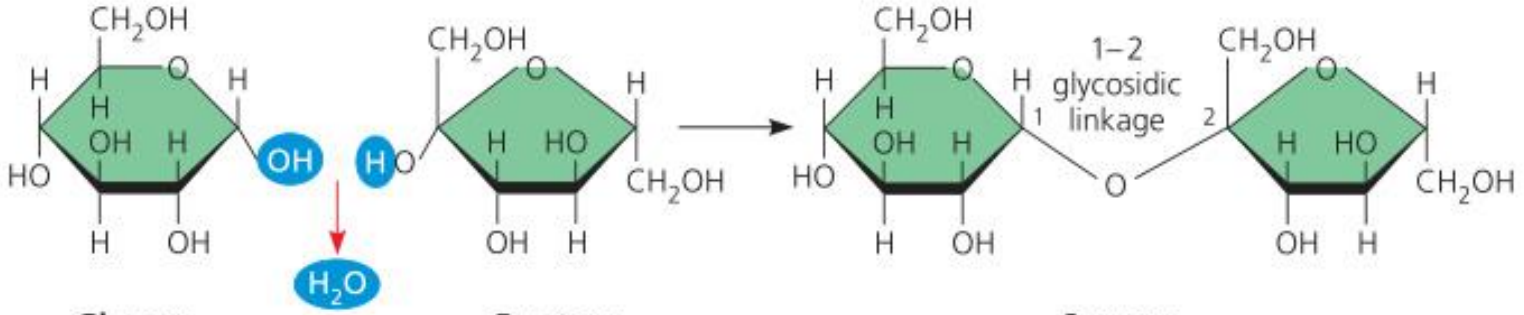
A **disaccharide** is formed when a dehydration reaction joins two monosaccharides by removing H_2O to form a covalent bond. This covalent bond is called a ***glycosidic linkage.***

▼ **Figure 5.5 Examples of disaccharide synthesis.**

(a) Dehydration reaction in the synthesis of maltose. The bonding of two glucose units forms maltose. The 1–4 glycosidic linkage joins the number 1 carbon of one glucose to the number 4 carbon of the second glucose. Joining the glucose monomers in a different way would result in a different disaccharide.



(b) Dehydration reaction in the synthesis of sucrose. Sucrose is a disaccharide formed from glucose and fructose. Notice that fructose forms a five-sided ring, though it is a hexose like glucose.



Polysaccharides

Polysaccharides, the polymers of sugars, have storage and structural roles.

Some polysaccharides serve as storage material, hydrolyzed as needed to provide monosaccharides for cells.

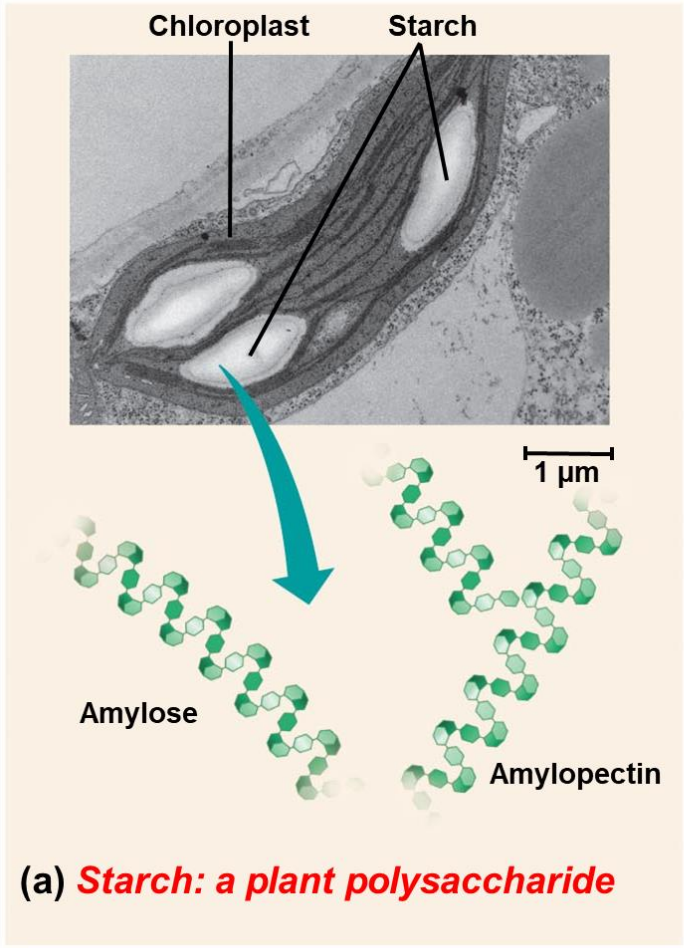
Other polysaccharides serve as building material for structures that protect the cell or the whole organism.

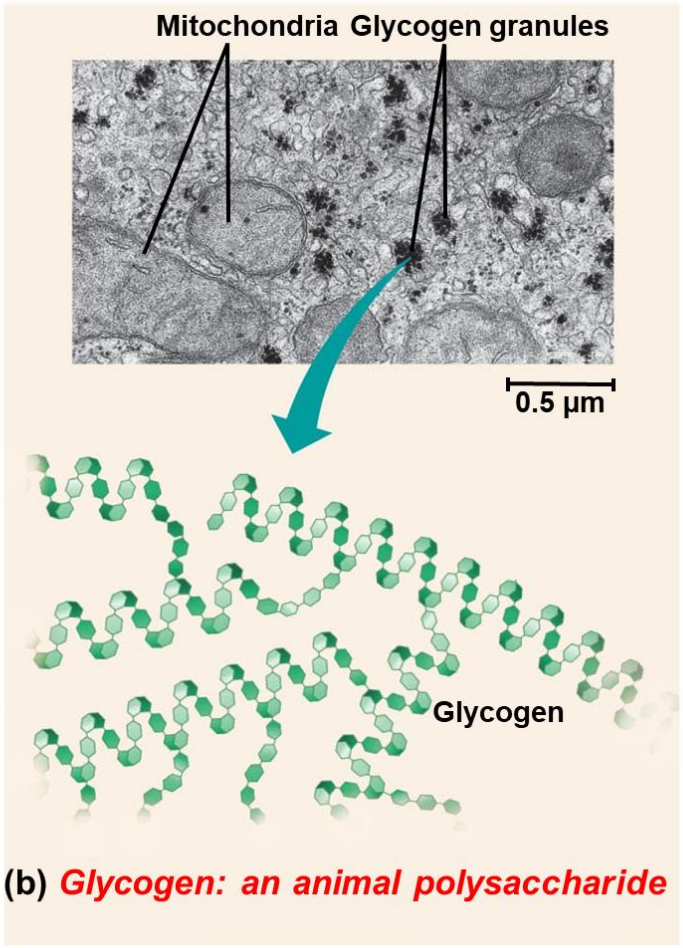
The structure and function of a polysaccharide are determined by its sugar monomers and the positions of glycosidic linkages.

Storage Polysaccharides

Starch is a plant storage polysaccharide.
Starch is made of glucose monomers.
Plants store surplus starch as granules within chloroplasts and other plastids.
• The simplest form of starch is **amylose**.

Glycogen is a storage polysaccharide in Animals.
• Humans and other vertebrates store glycogen mainly in liver and muscle cells



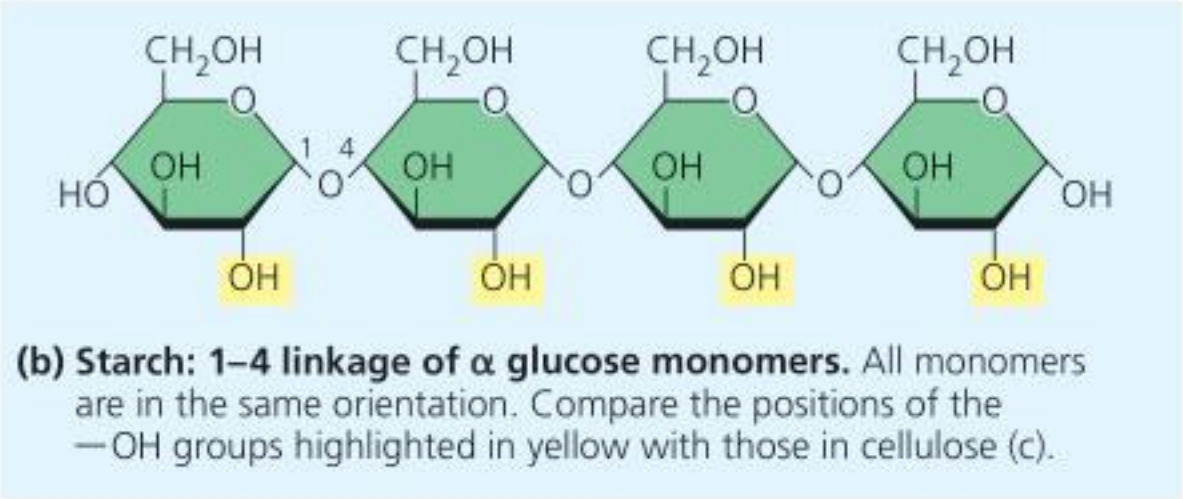


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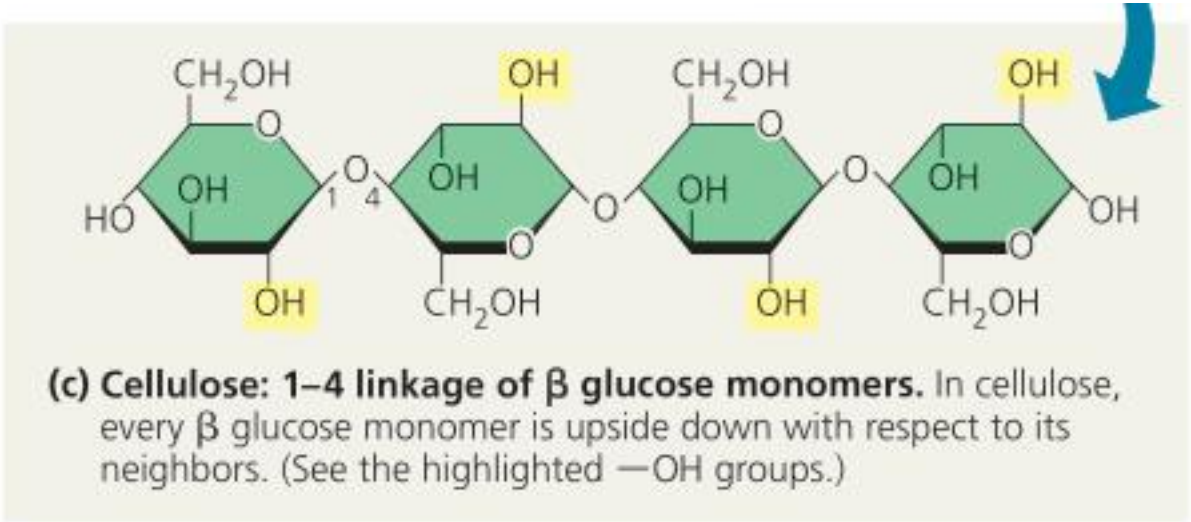
Structural Polysaccharides

The polysaccharide cellulose is a major component of plant cell walls.
Like starch, cellulose is a polymer of glucose, but the glycosidic linkages differ.
The difference is based on two ring forms for glucose: alpha (α) and beta (β)

- Polymers with α glucose are helical.
- Polymers with β glucose are straight.



(b) Starch: 1–4 linkage of α glucose monomers. All monomers are in the same orientation. Compare the positions of the —OH groups highlighted in yellow with those in cellulose (c).



(c) Cellulose: 1–4 linkage of β glucose monomers. In cellulose, every β glucose monomer is upside down with respect to its neighbors. (See the highlighted —OH groups.)

- In straight structures, H atoms on one strand can bond with OH groups on other strands.
- Parallel cellulose molecules held together this way are grouped into microfibrils, which form strong building materials for plants.

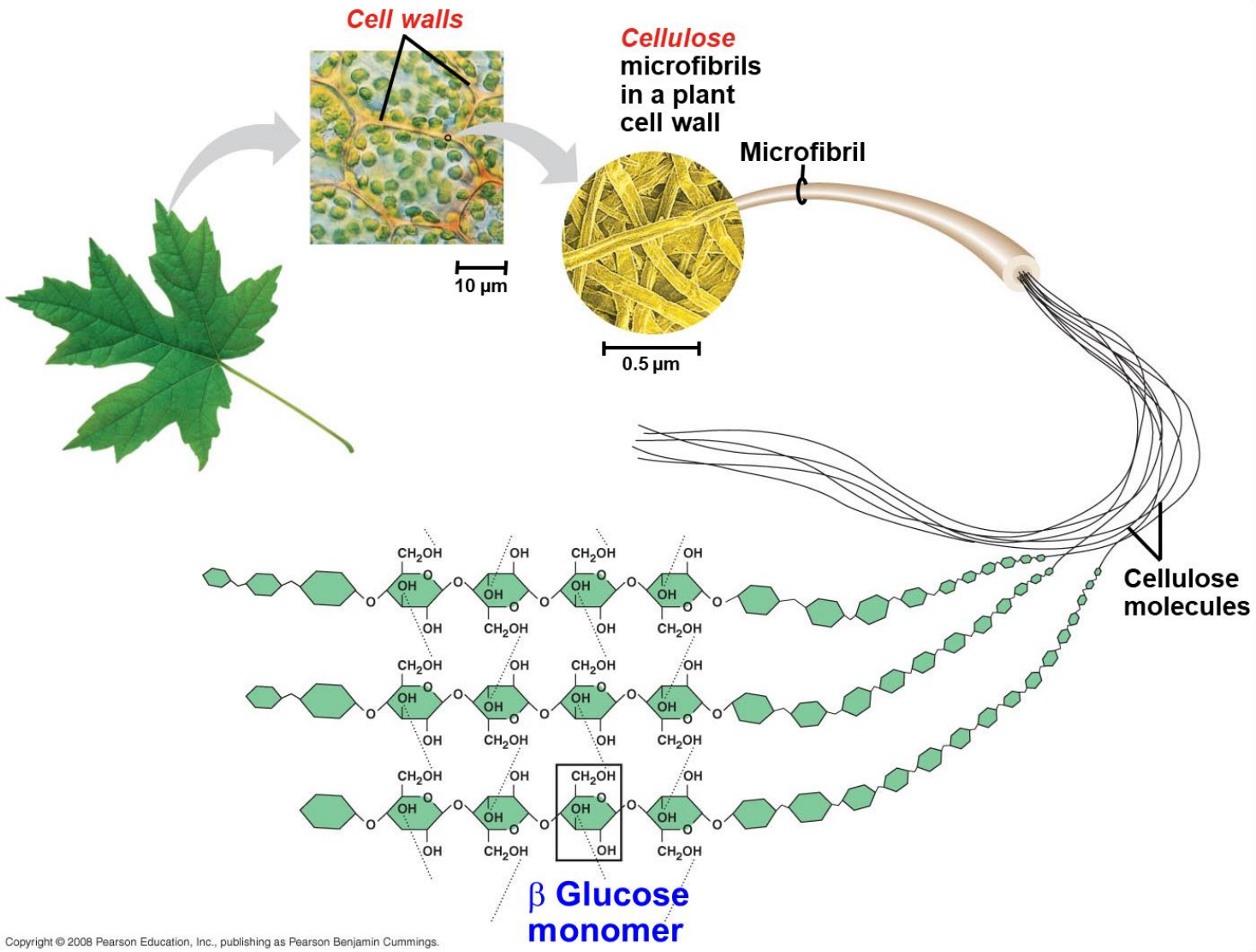
Structural Polysaccharides

Enzymes that digest starch by hydrolyzing α linkages can't hydrolyze β linkages in cellulose.

Animals can't digest cellulose. Therefore, cellulose in human food passes through the digestive tract as **insoluble fiber**.

Some microbes (ruminococcus) use enzymes (cellulase) to digest cellulose.

Many herbivores, from cows to termites, have symbiotic relationships with these microbes..



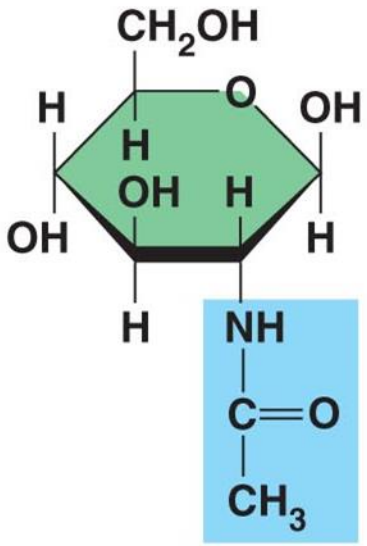
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Structural Polysaccharides

Chitin is another important structural polysaccharide.

Chitin provides structural support for the cell walls of fungi and also found in the exoskeletons (hard cases that surround the soft parts of an animal) of arthropods (insects, spiders, crustaceans, and related animals)

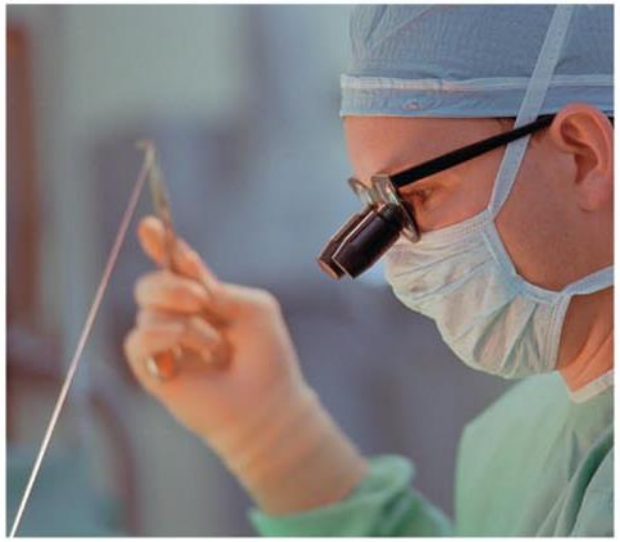
Chitin is similar to cellulose, with β linkages, except that the glucose monomer of chitin has a nitrogen-containing attachment.



(a) The structure of the *chitin monomer.*



(b) Chitin forms the *exoskeleton* of arthropods.



(c) Chitin is used to make a strong and flexible surgical thread.

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